



ANSYS LAB 1

1-D Element Analysis

Machine Construction

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Sec 2
Team 4

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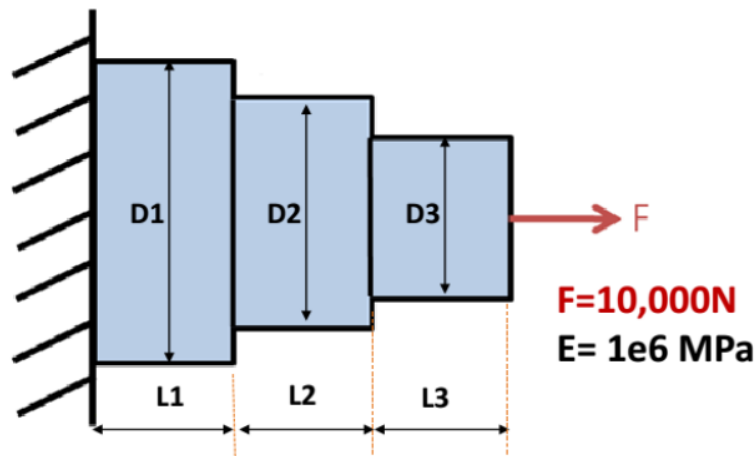
Objectives

- Knowledge and understanding of finite element analysis of 1D Elements
- Ability to use Ansys Mechanical to set up a 1D structural simulation
- Awareness of the importance of simulation in engineering design

Note: each dimension and force is multiplied by 5 (team 4 + 1).

Solutions

Problem 1

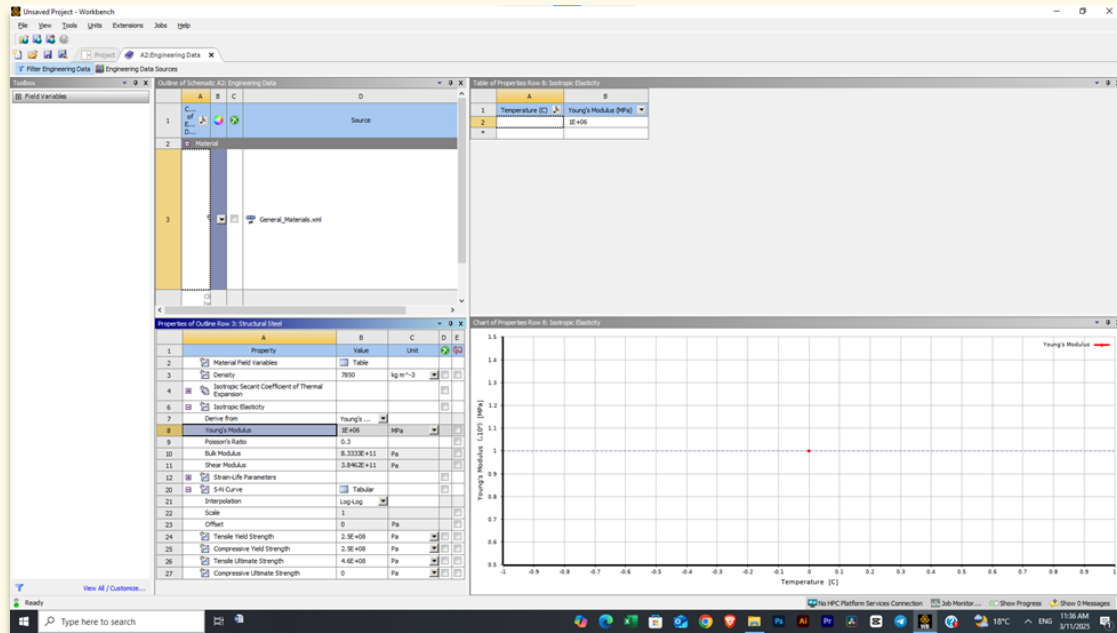


$L1=L2=L3=40$
 $D1=50, D2=40, D3=30$

Using Workbench

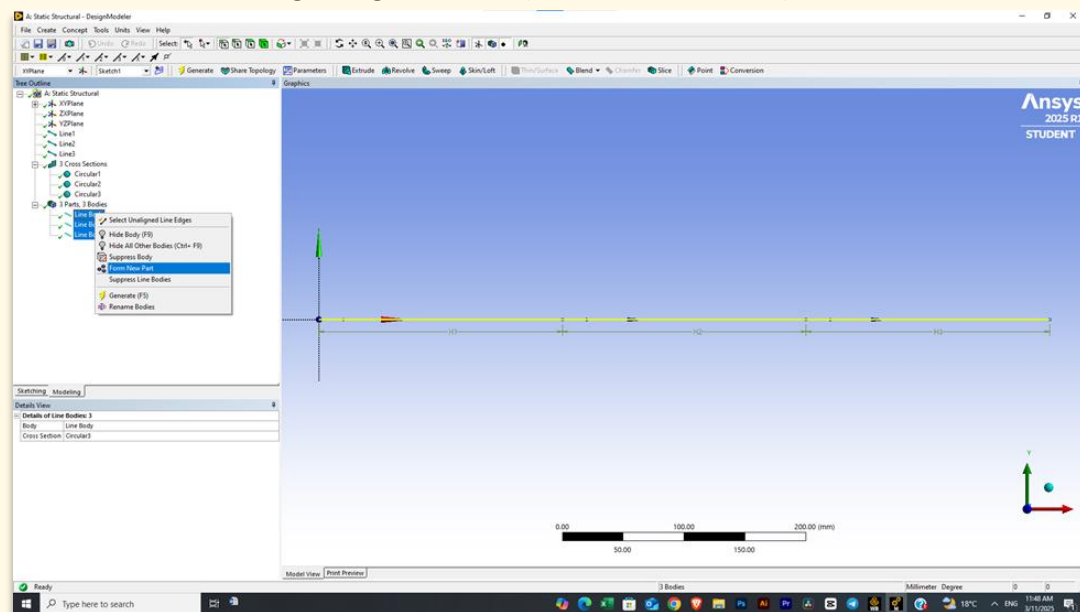
Description

Change Young's Modulus value to 1E6 Pa.



Description

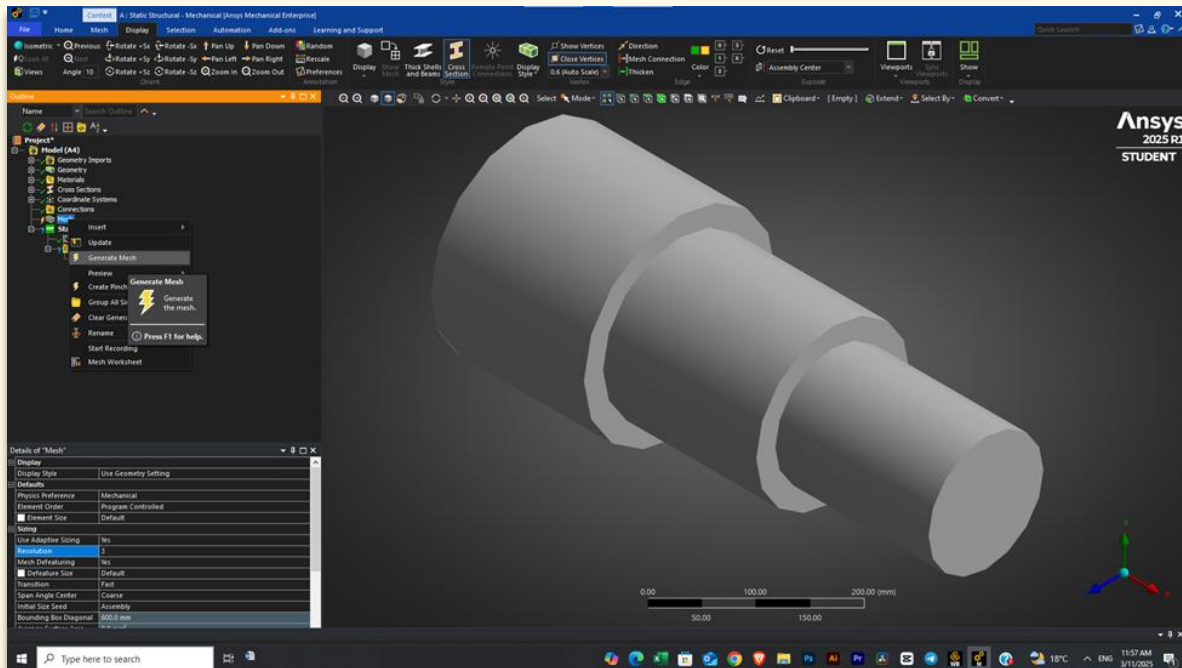
Draw structure using design modeler, add cross-section, and form a new part.



Step 3: Preprocessing – Editing Model – Adjusting Mesh

Description

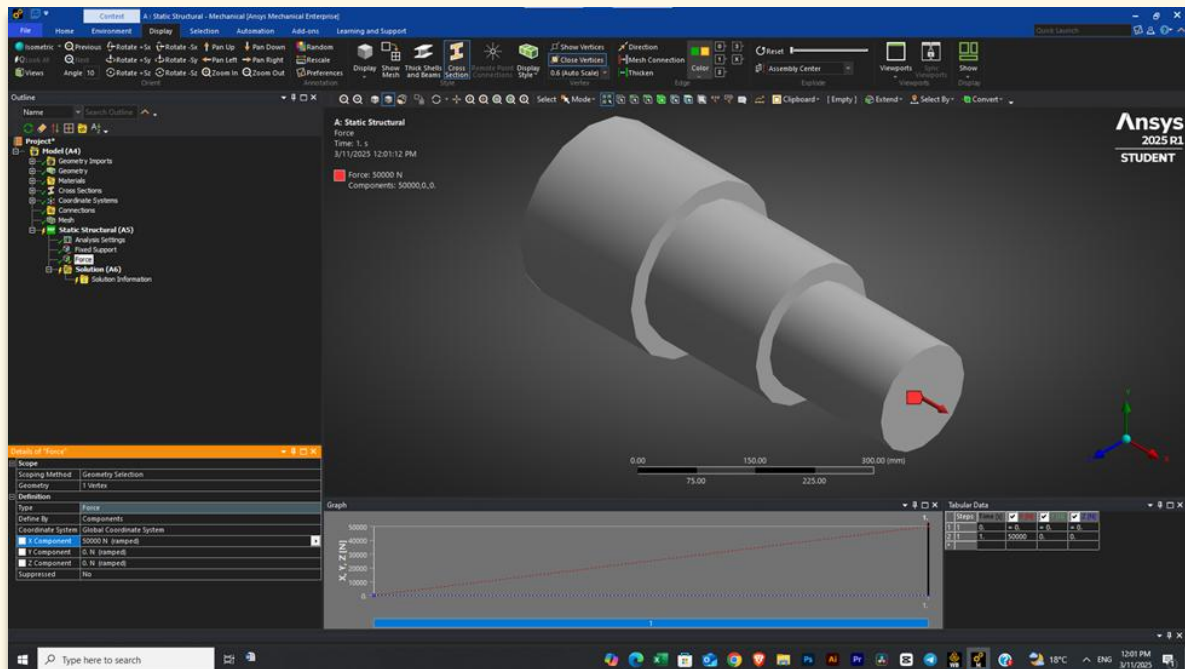
Choosing an appropriate initial **mesh resolution** for *model*, 5.



Step 4: Preprocessing – Editing Model – Boundary Conditions

Description

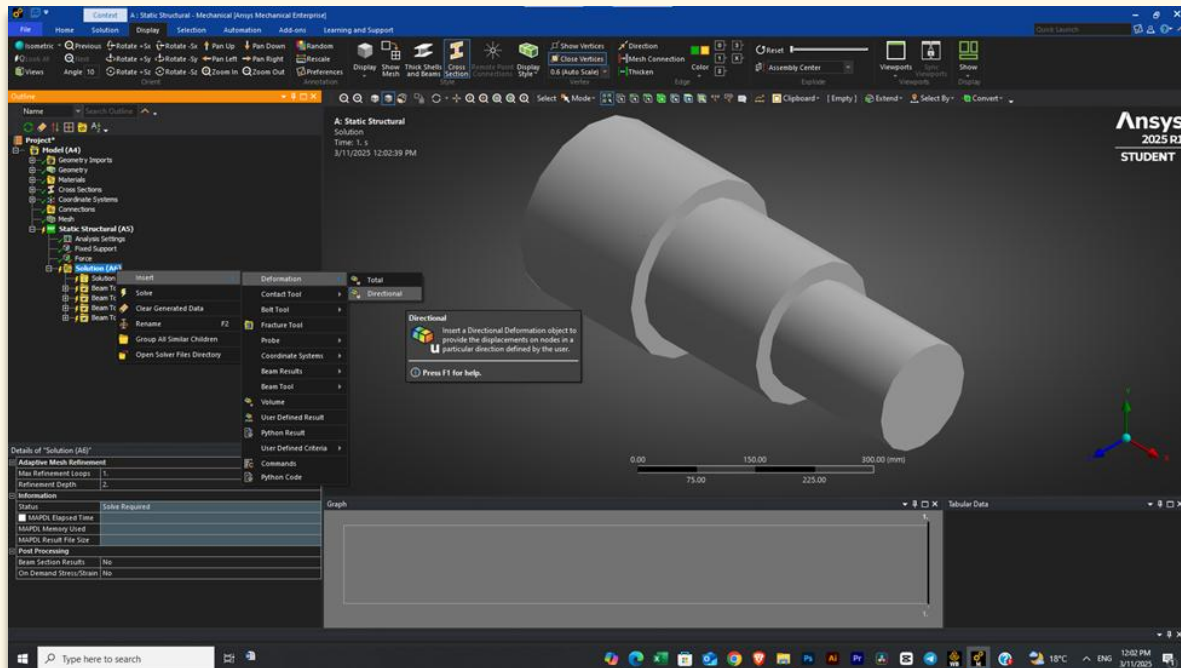
Insert **forces** and **fixation** to *structure*.



Step 5: Postprocessing – Directional Deformation

Description

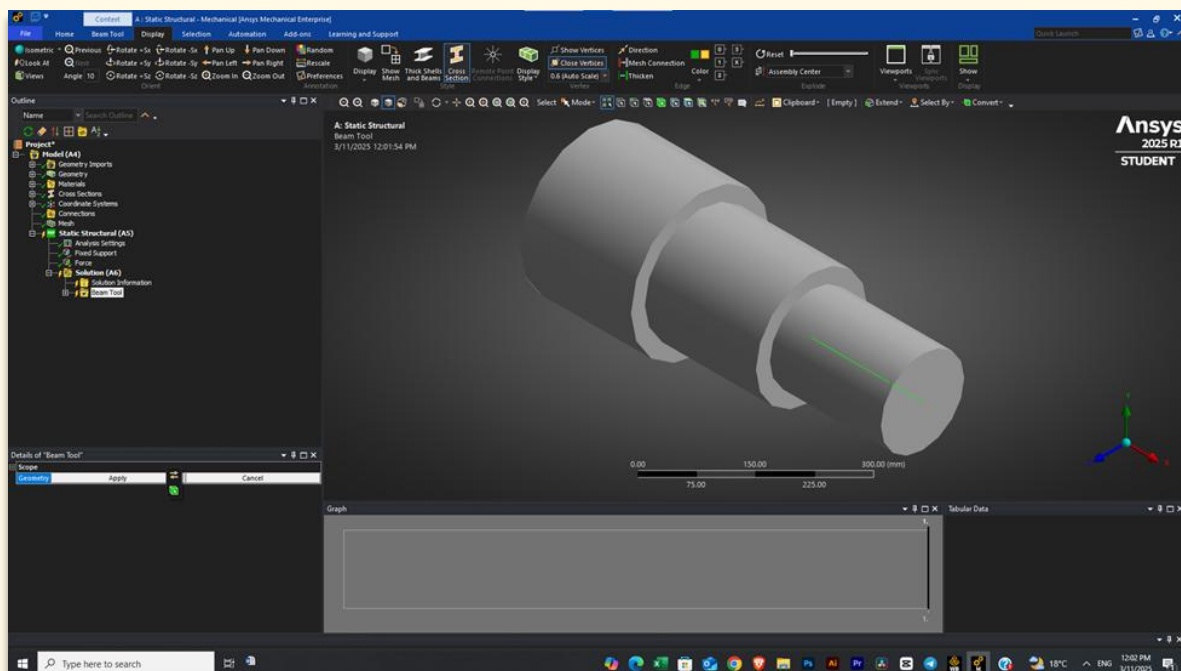
Insert **directional deformation** to *solution* and choose **all body** as *scope geometry*.



Step 6: Postprocessing – Stress Measurement

Description

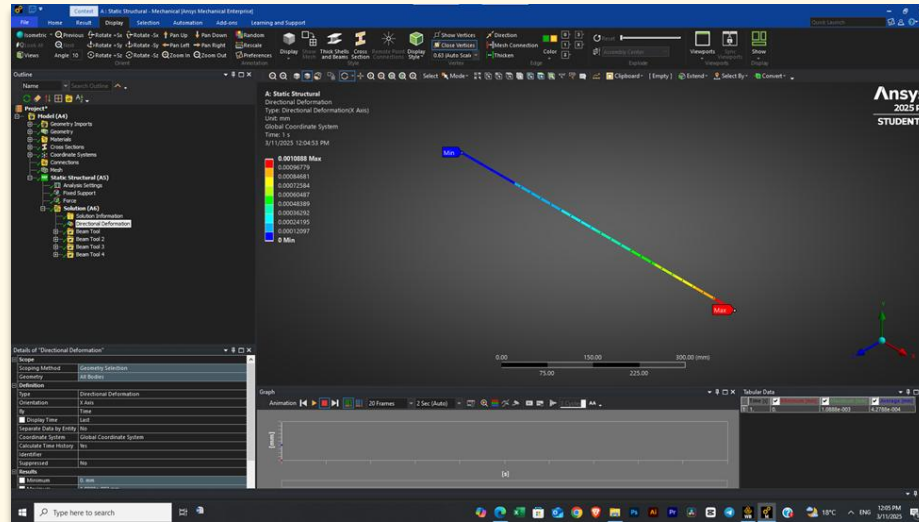
Insert **beam tool** for each beam to *solution* and choose the beams of the structure as *scope geometry*.



Step 7: Postprocessing – Deformation

Description

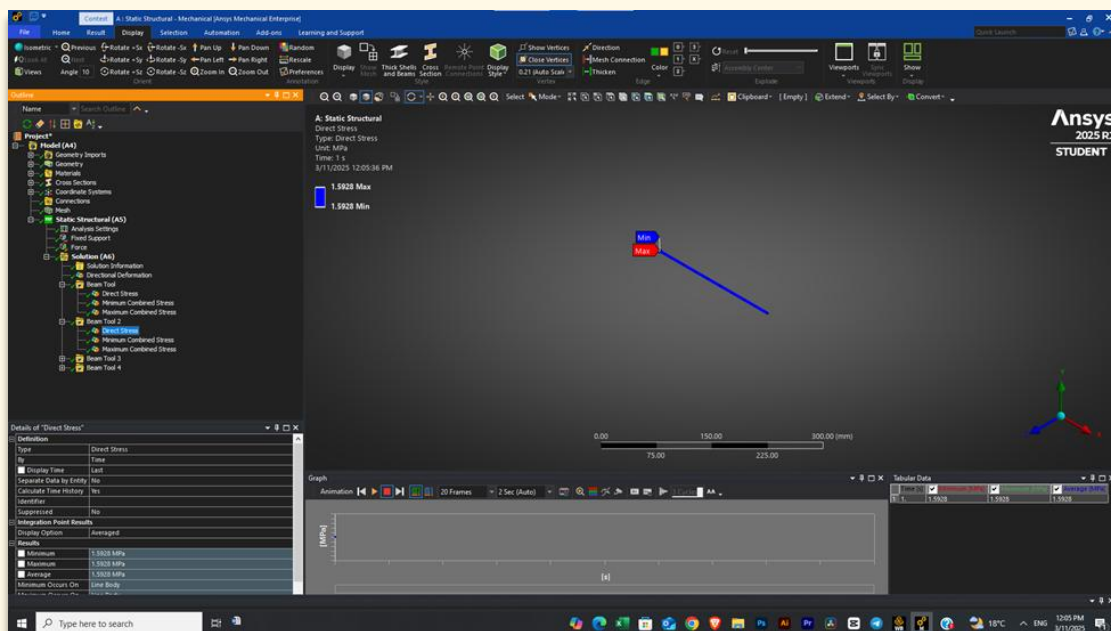
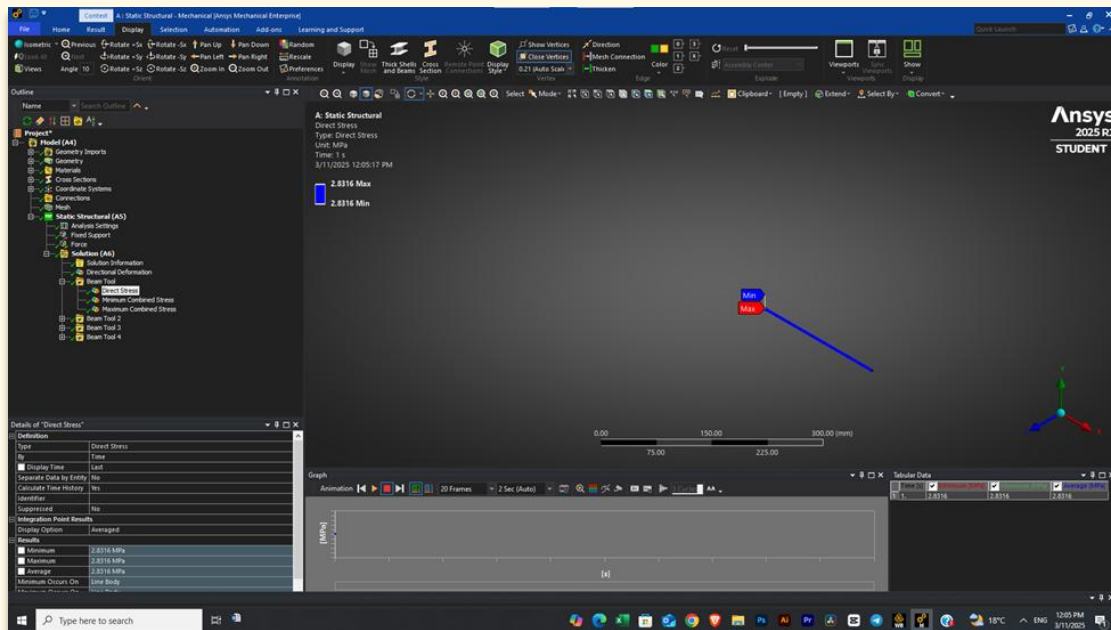
Press on solve and get directional deformation (x-axis).



Step 7: Postprocessing – Stresses

Description

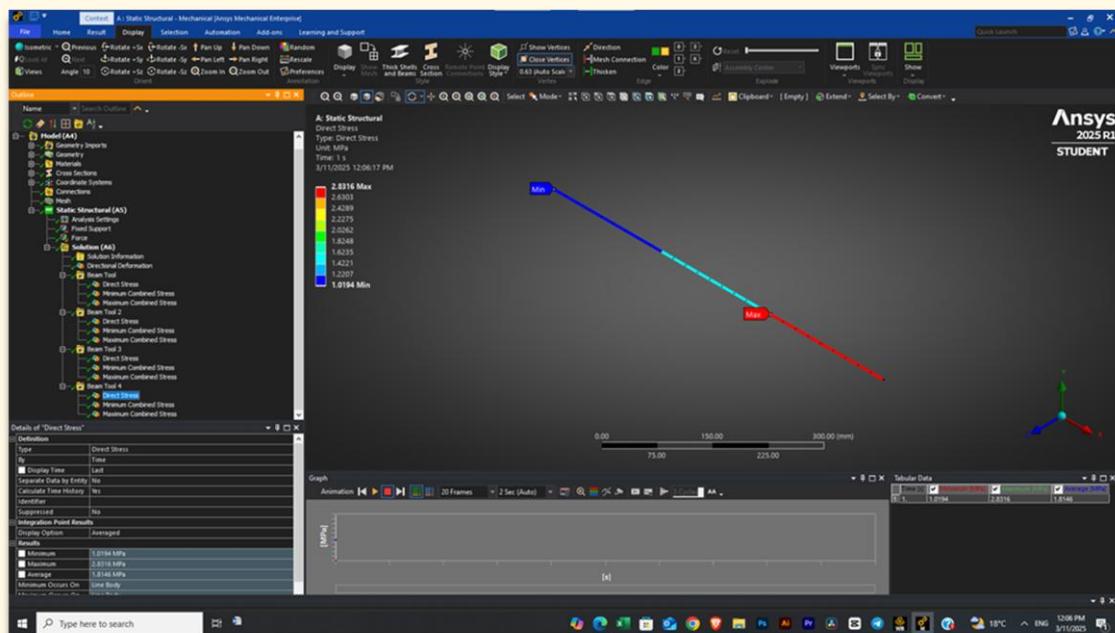
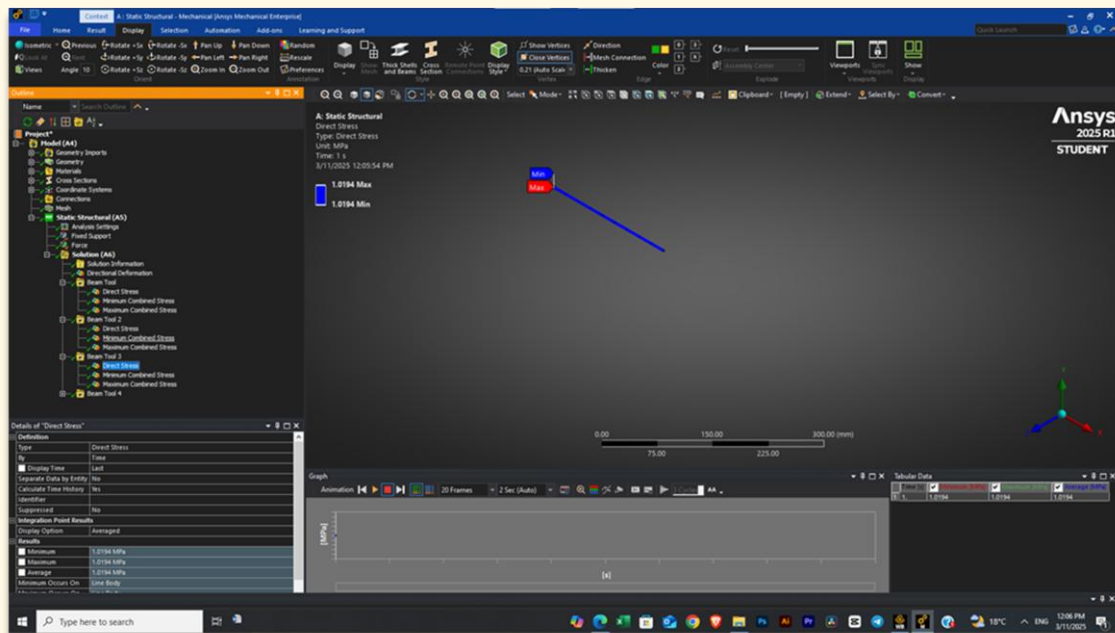
Get stress on each beam.



Step 7: Postprocessing – Stresses

Description

Get stress on each beam.



Step 8: Postprocessing – Tabulate Results

Description

Put results in a table.

Results

Directional Deformation (x)	[-0.00mm, -0.0011mm]
Stress on Bar 1	2.83 MPa
Stress on Bar 2	1.59 MPa
Stress on Bar 3	1.02

Using APDL

APDL SCRIPT Steps

Description

- Preprocessing
 - Engineering Data: adjusting real constants and material properties
 - Model: adding nodal locations, element connections, nodal constraints, and loads
- Postprocessing
 - Plotting model
 - Listing nodal displacements, element stresses, element forces.

APDL SCRIPT

```
/filnam, 01 Rep1
/title, Stepped Bar
/prep

!elements
et,1,link1

!real constants
r,1,49087.38
r,2,31415.92
r,3,17671.49

!material properties
mp,ex,1,30e6

!nodal locations
n,1,0,0
n,2,200,0
n,3,400,0
n,4,600,0

!element connections
real,1
en,1,1,2

real,2
en,2,2,3

real,3
en,3,3,4

!nodal constraints
d,1,all,0
```

APDL SCRIPT

```
!loading
f,4,fx,500000

finish

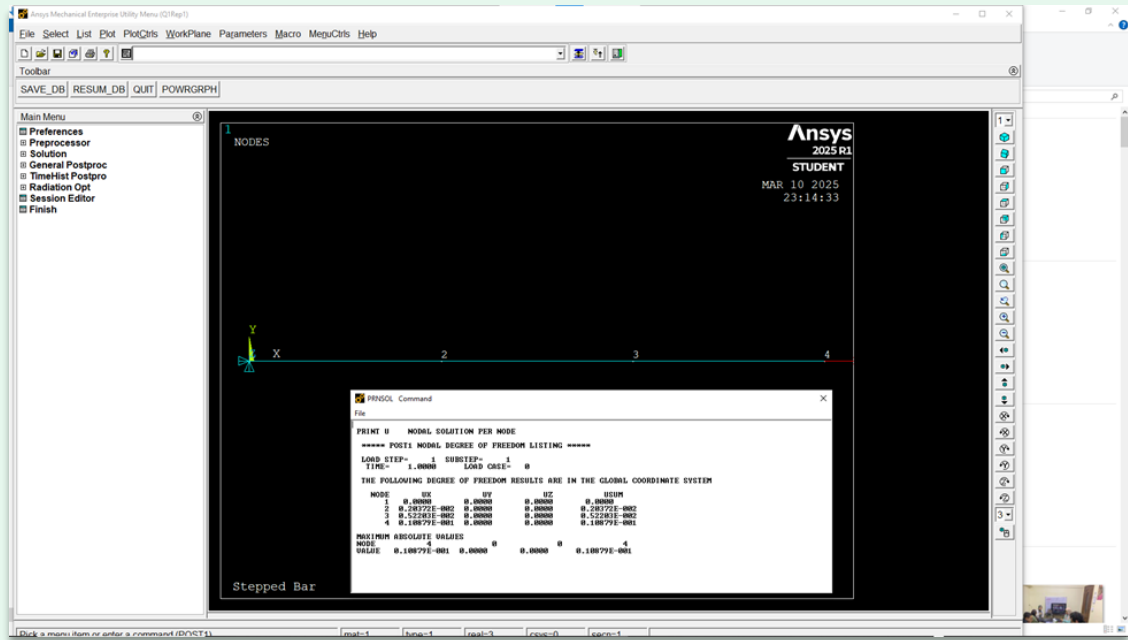
/solu
antype,static
solve
save
finish

/post1

etable,Force,SMISC,1
etable,Stress,ls,1

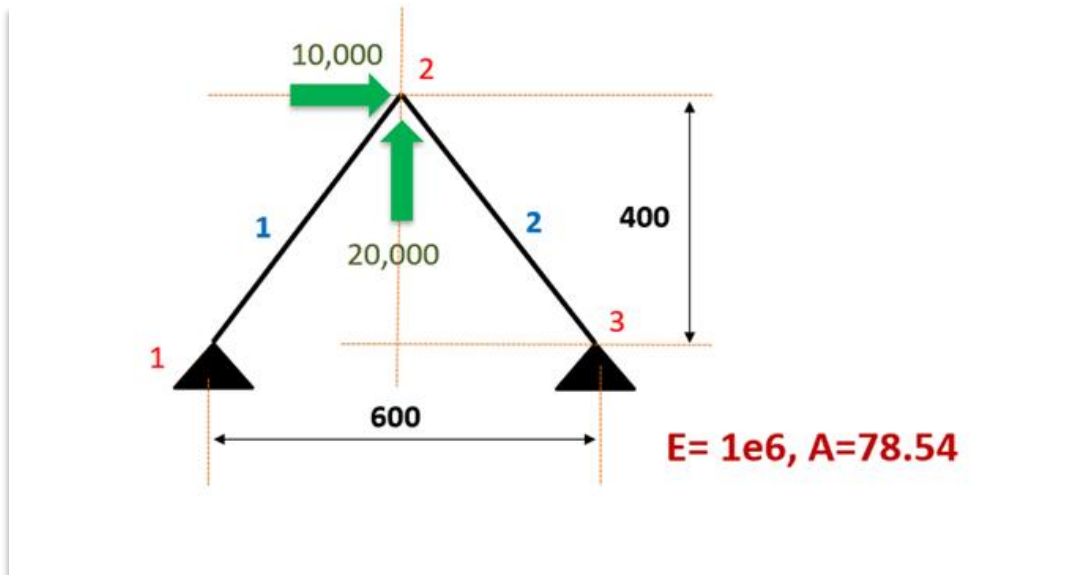
/output,resultsfile,dat
prnsol,u
prtab,Force,Stress

/output
```



NODE	UX	UY	UZ	USUM
1	0	0	0	0
2	2.04E-03	0	0	2.04E-03
3	5.22E-03	0	0	5.22E-03
4	1.09E-02	0	0	1.09E-02

Problem 2

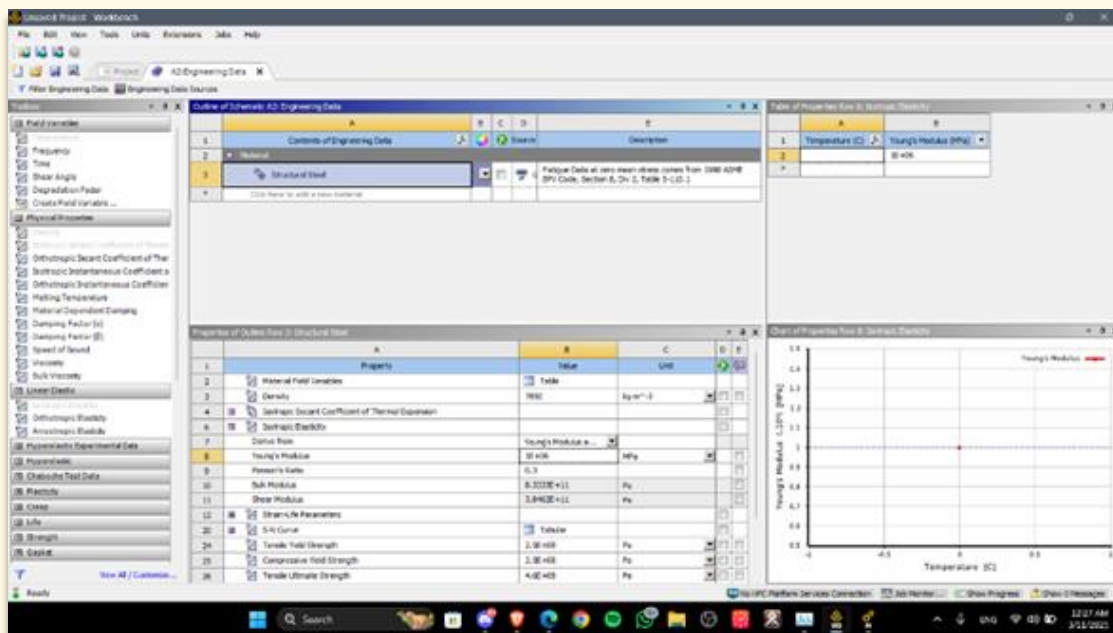


Using Workbench

Step 1: Preprocessing – Editing Engineering Data

Description

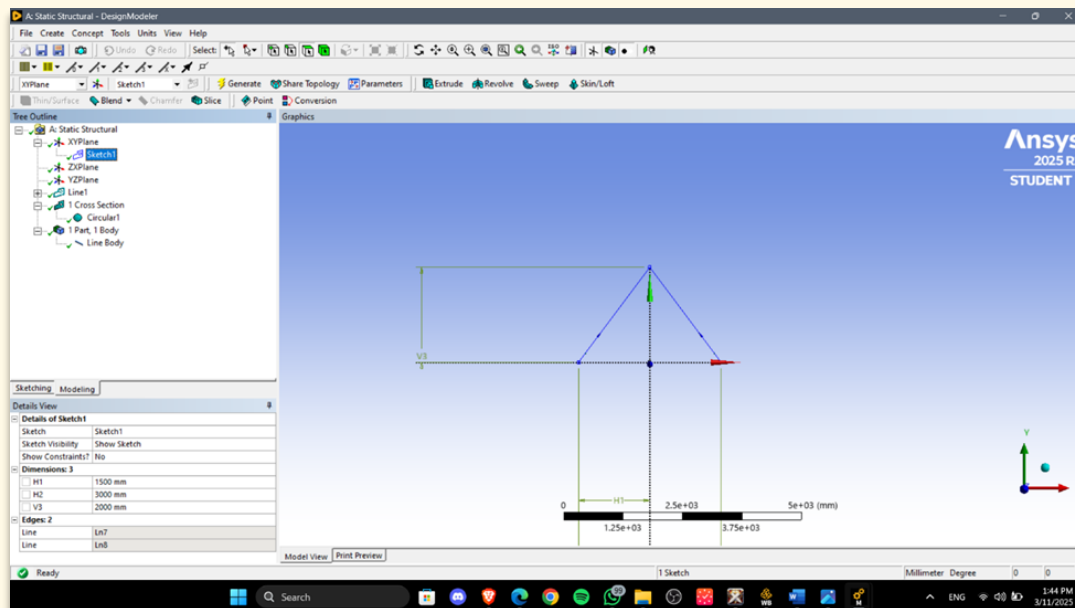
Change Young's Modulus value to 1E6 Pa.



Step 2: Preprocessing – Editing Geometry

Description

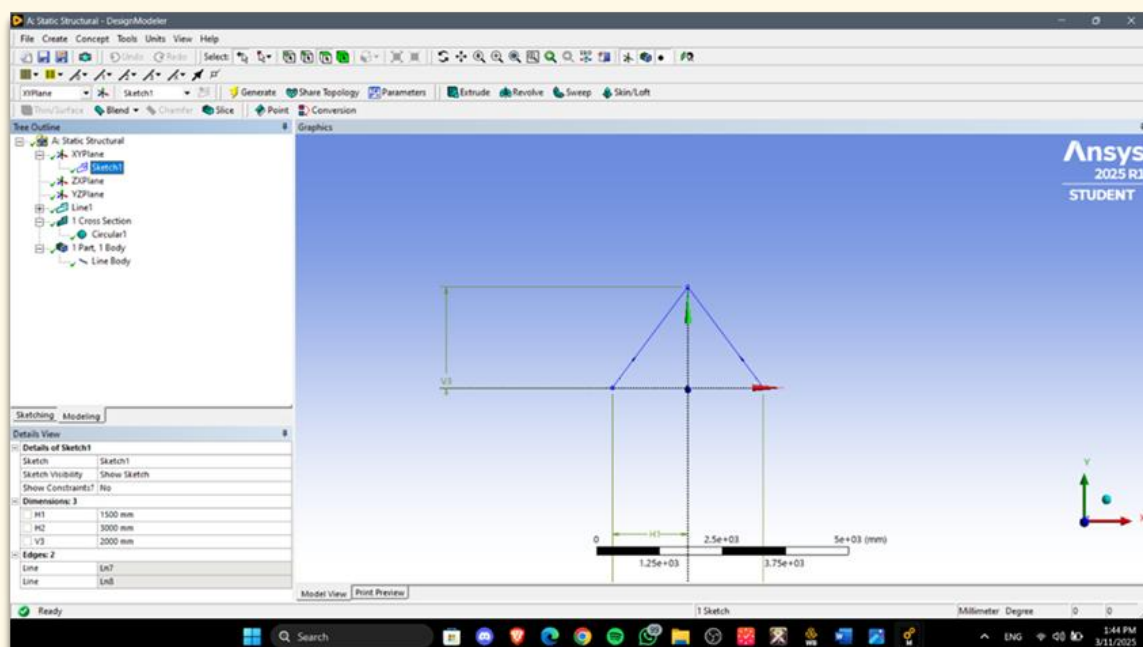
Draw structure using design modeler and add cross-section.



Step 3: Preprocessing – Editing Model – Adjusting Mesh

Description

Choosing an appropriate initial **mesh resolution** for *model*, 5.

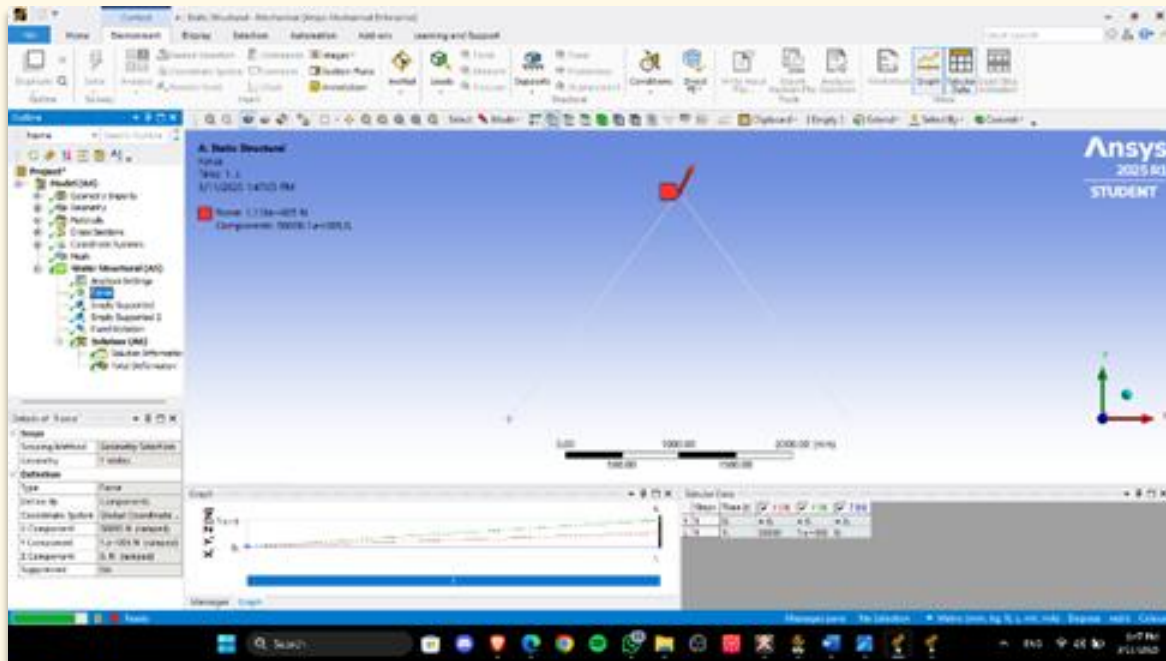


Step 4: Preprocessing – Editing Model – Boundary Conditions

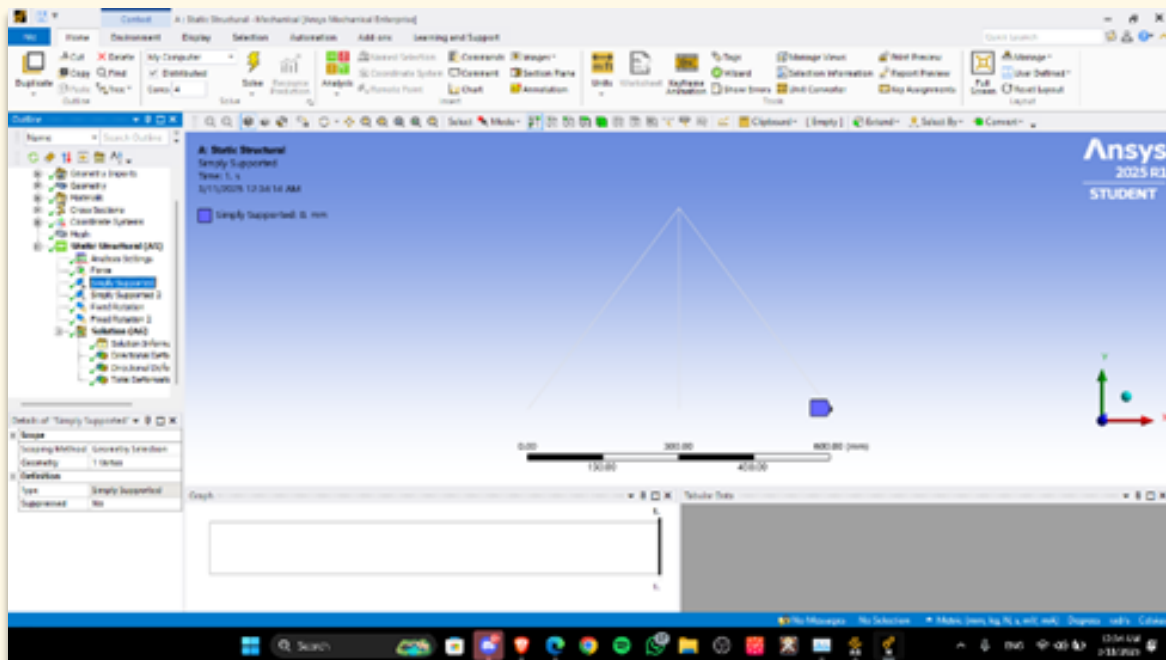
Description

Insert **forces** and **fixation** to *structure*.

Force:



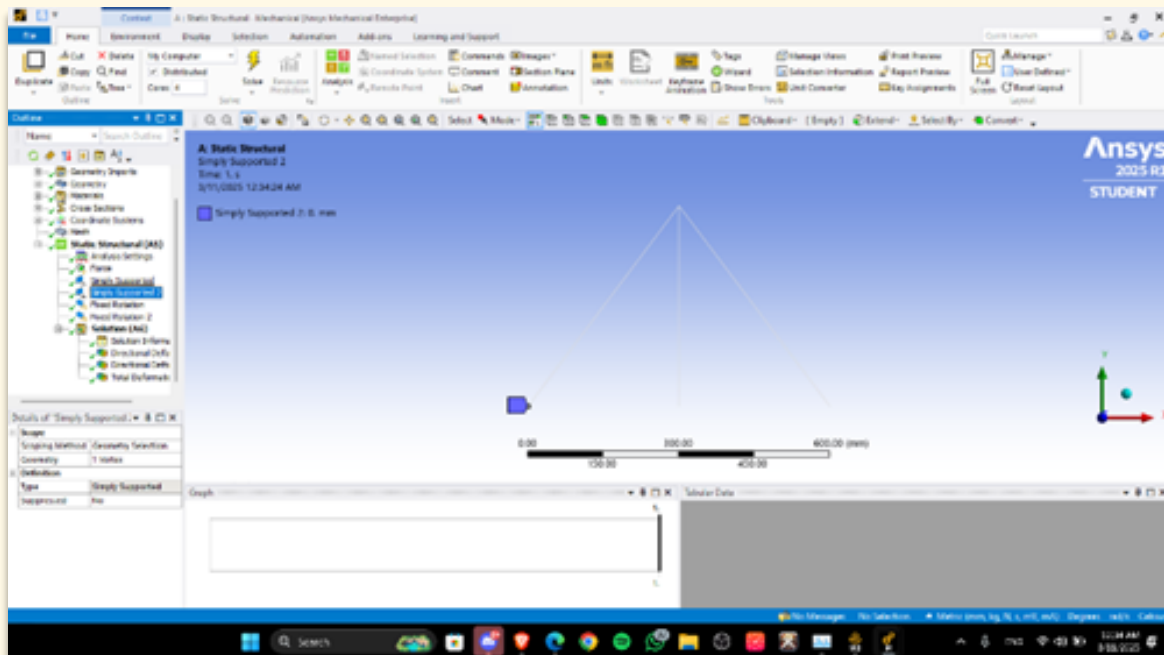
Simple Support:



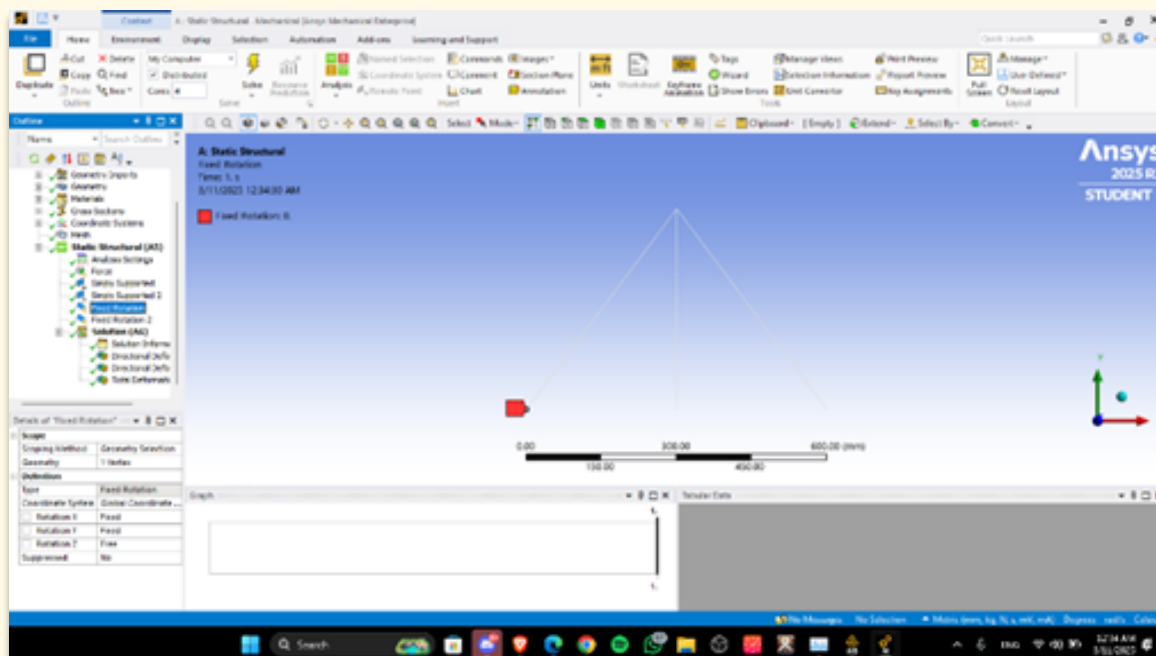
Description

Insert **forces** and **fixation** to *structure*.

Simple Support:



Fixed rotation:



Step 5: Postprocessing – Directional Deformation

Description

Insert **directional deformation** to *solution* and choose **all body** as *scope geometry*.

- Right click then “**Insert**” and then “**Deformation**” then “**one dim on x axis/ one dim on y axis/Total**”

Step 6: Postprocessing – Stress Measurement

Description

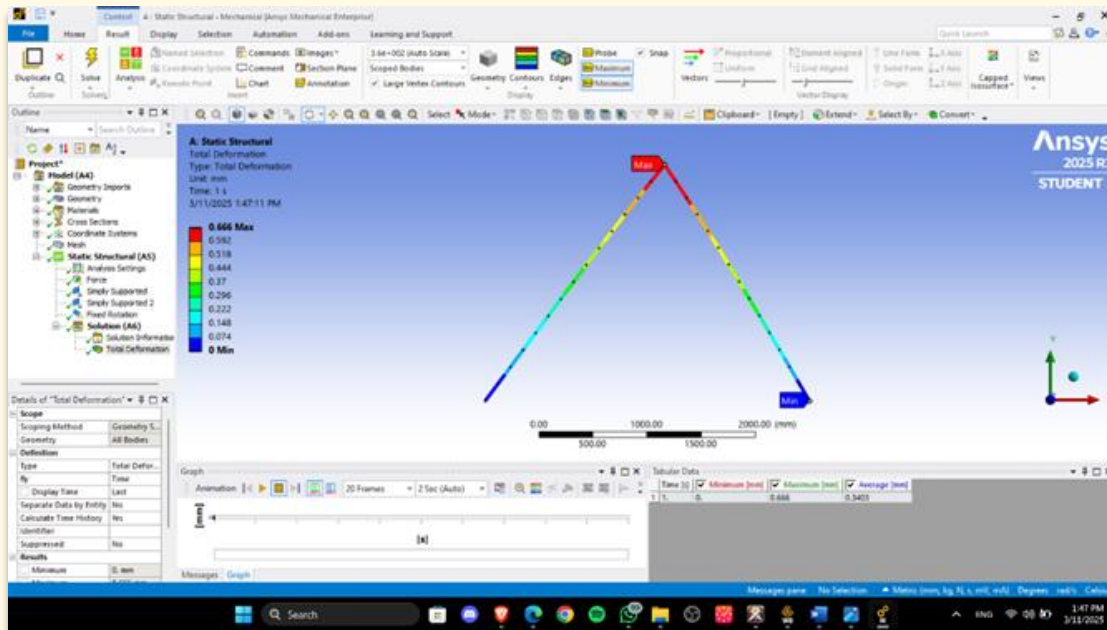
Insert **beam tool** for each beam to *solution* and choose the beams of the structure as *scope geometry*.

- Right Click on “**Solution**”, then “**Insert**” and Select “**Beam Tool**”
- Select a section from that beam - Go to “Geometry” at the details table and select “**Apply**”
- Select another beam tools for the remaining member

Step 7: Postprocessing – Deformation

Description

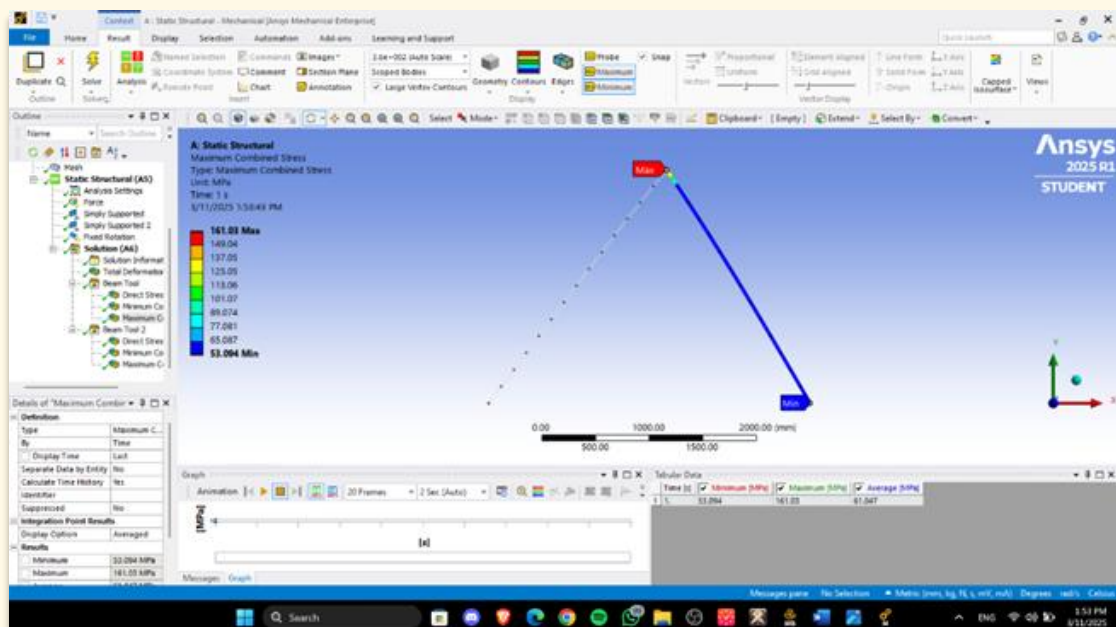
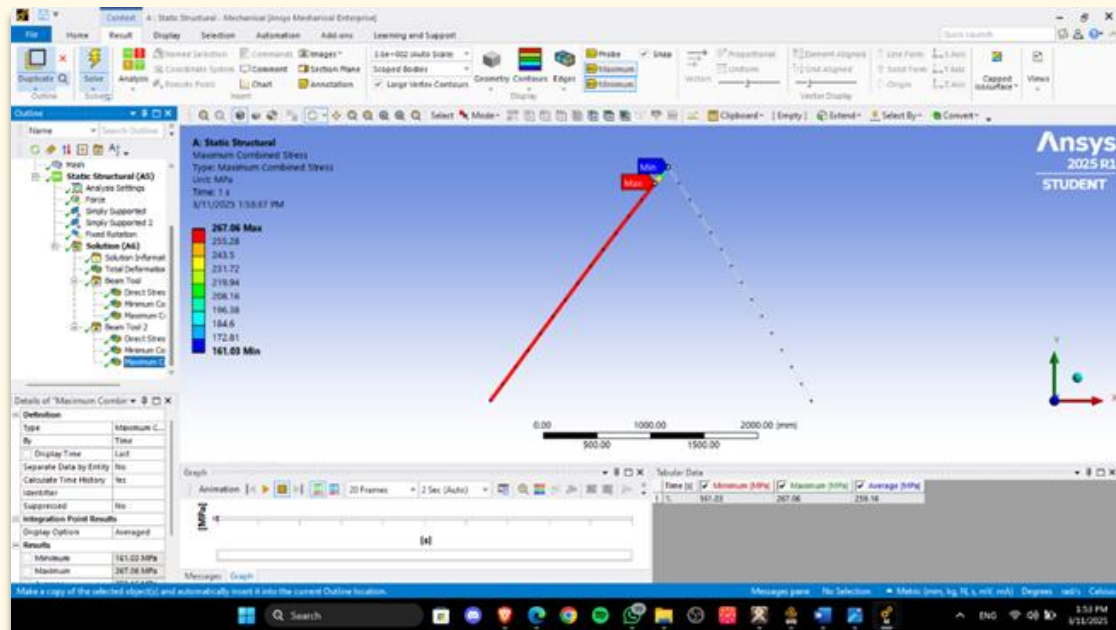
Press on solve and get total deformation.



Step 7: Postprocessing – Stresses

Description

Get stress on each beam.



Step 8: Postprocessing – Tabulate Results

Description

Put results in a table.

Results

Total Deformation	[0.00mm, 0.67 mm]
Stress on Bar 1 (1-2)	[161.03 MPa, 267.06 MPa]
Stress on Bar 2 (2-3)	[53.09 MPa, 161.03 MPa]

Using APDL

APDL SCRIPT Steps

Description

- Preprocessing
 - Engineering Data: adjusting real constants and material properties
 - Model: adding nodal locations, element connections, nodal constraints, and loads
- Postprocessing
 - Plotting model
 - Listing nodal displacements, element stresses, element forces.

APDL SCRIPT

```
/filenam,Q2_Truss
/title,Q2 Truss
/prep7

et,1,link1

r,1,78.54

mp,ex,1,1e6

n,1,0,0,0
n,2,1500,2000,0
n,3,3000,0, 0

Real,1
en,1,1,2
en,2,2,3

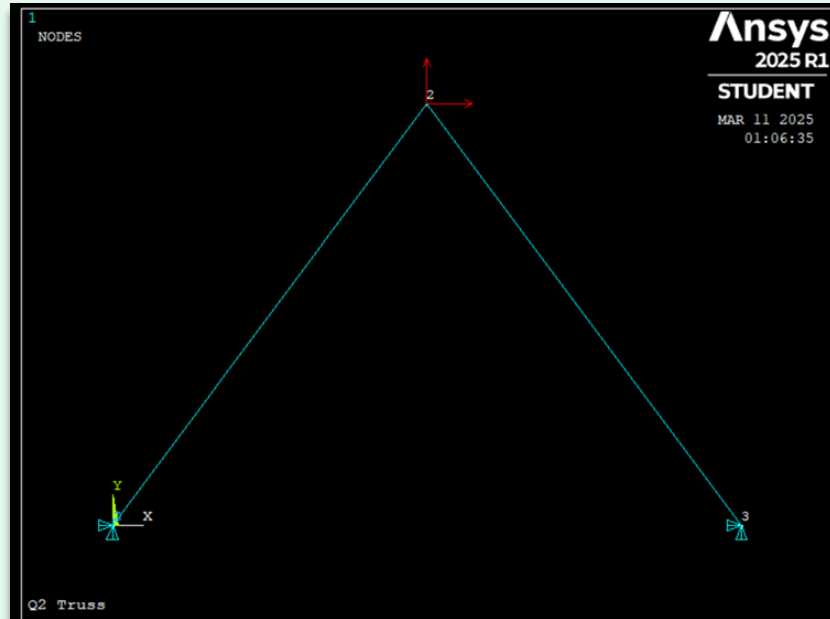
d,1,ux,0
d,1,uy,0
d,3,ux,0
d,3,uy,0
f,2,fx,20000
f,2,fy,100000

finish
/solu
antype,static
solve
save
finish

/post1
etable,Force,SMISC,1
etable,Stress,LS,1

/output,resultsfile,dot
prnsol,u
pretab,Force,Stress
/output
```

Results



PRNSOL Command

File

PRINT U NODAL SOLUTION PER NODE

***** POST1 NODAL DEGREE OF FREEDOM LISTING *****

LOAD STEP= 1 SUBSTEP= 1
TIME= 1.0000 LOAD CASE= 0

THE FOLLOWING DEGREE OF FREEDOM RESULTS ARE IN THE GLOBAL COORDINATE SYSTEM

NODE	UX	UY	UZ	USUM
1	0.0000	0.0000	0.0000	0.0000
2	0.88419	2.4868	0.0000	2.6393
3	0.0000	0.0000	0.0000	0.0000

MAXIMUM ABSOLUTE VALUES				
NODE	UX	UY	UZ	USUM
2	0.88419	2.4868	0.0000	2.6393

PRETAB Command

File

PRINT ELEMENT TABLE ITEMS PER ELEMENT

***** POST1 ELEMENT TABLE LISTING *****

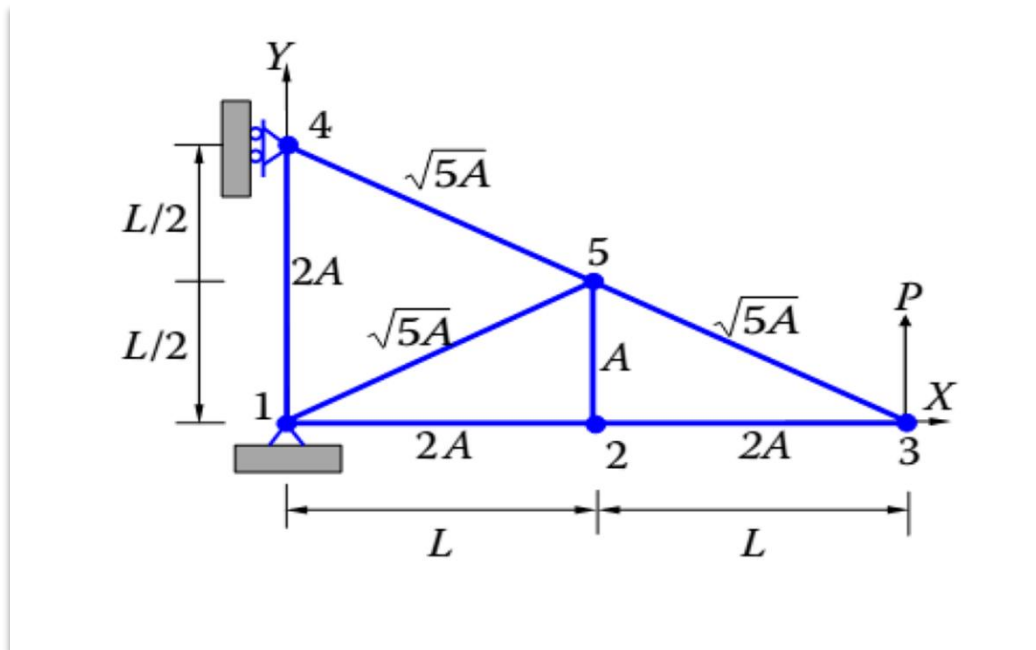
STAT	CURRENT	CURRENT
ELEM	FORCE	STRESS
1	79167.	0.0000
2	45833.	0.0000

MINIMUM VALUES		
ELEM	FORCE	STRESS
2	45833.	0.0000

MAXIMUM VALUES		
ELEM	FORCE	STRESS
1	79167.	0.0000

NODE	UX	UY	UZ	USUM
1	0	0	0	0
2	0.88419	2.4868	0	2.6393
3	0	0	0	0

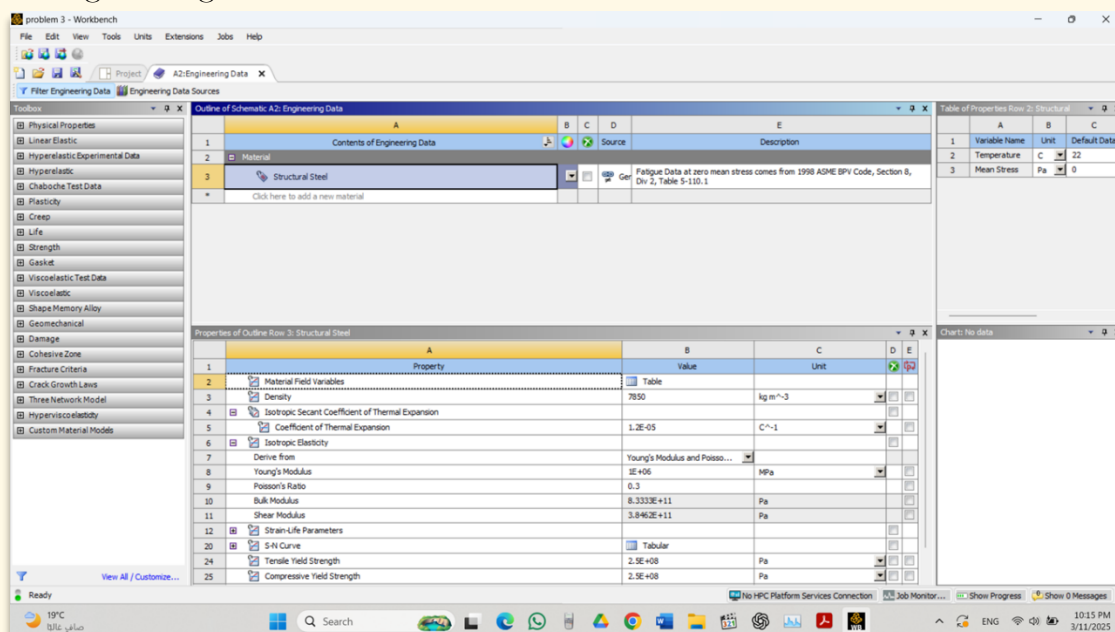
Problem 3



Using Workbench

Description

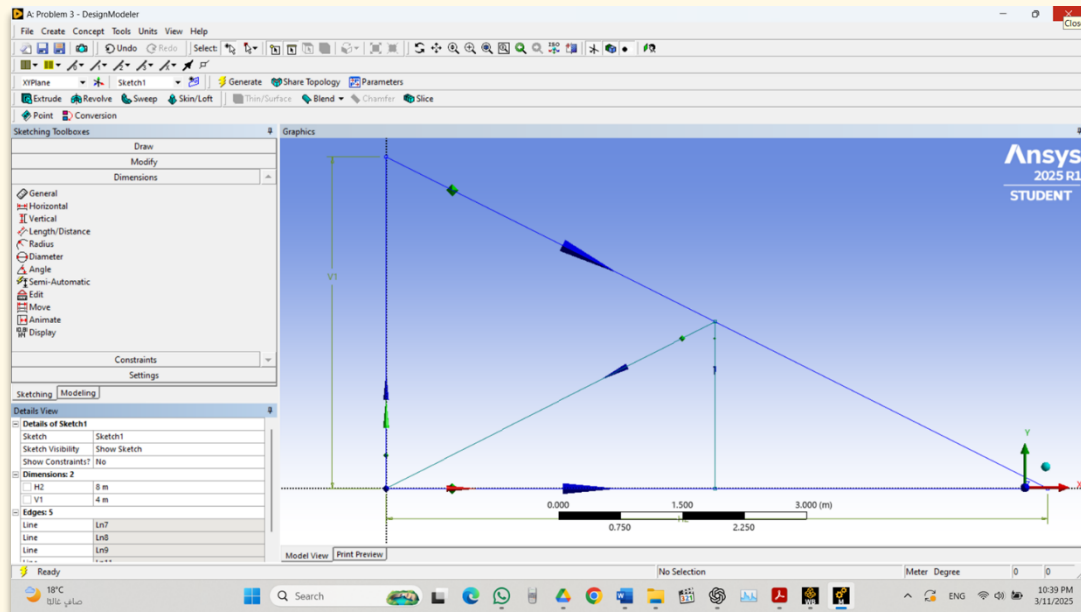
Change Young's Modulus value to 1E6 Pa.



Step 2: Preprocessing – Editing Geometry

Description

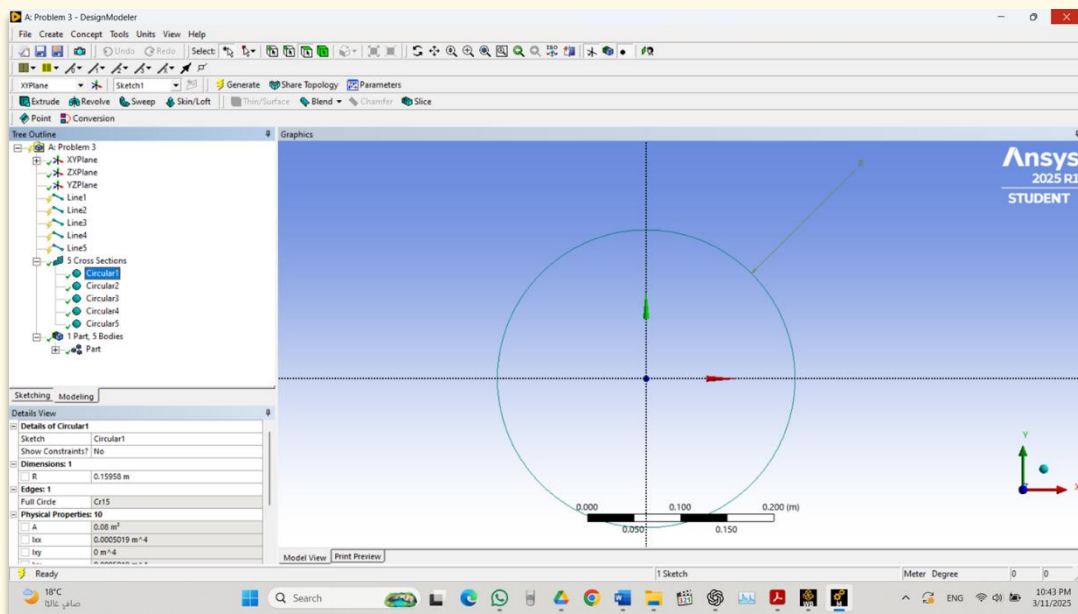
Draw structure using design modeler.



Step 2: Preprocessing – Editing Geometry

Description

Add cross-sections.



Step 3: Preprocessing – Editing Model – Adjusting Mesh

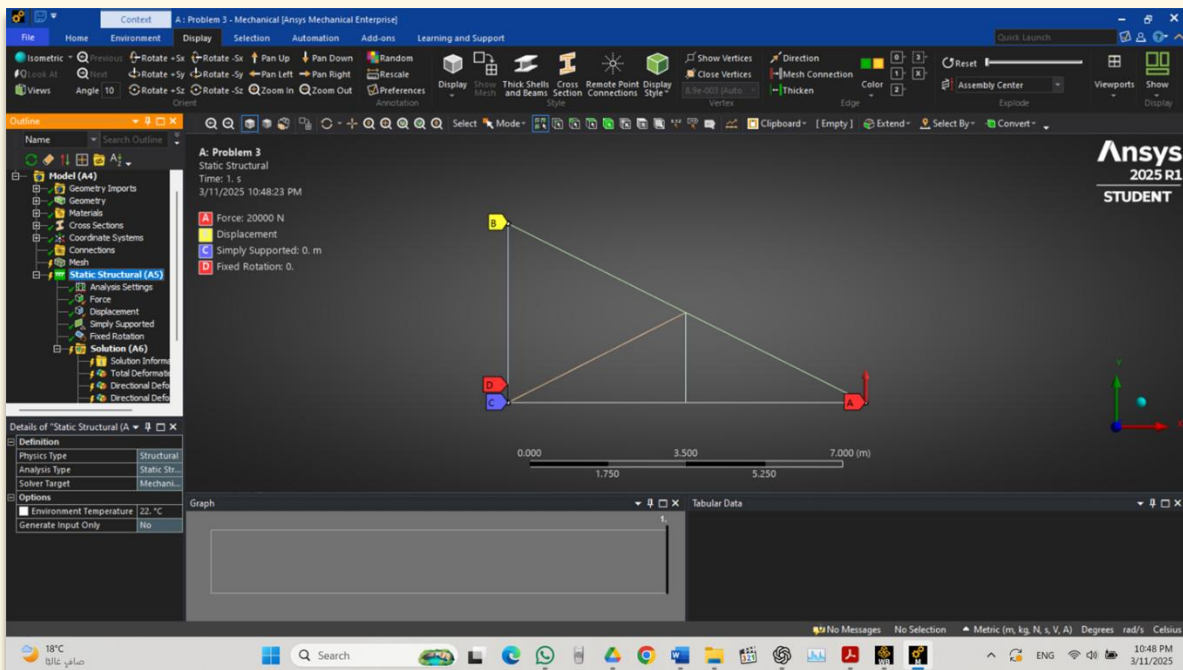
Description

Choosing an appropriate initial **mesh resolution** for *model*, 5.

Step 4: Preprocessing – Editing Model – Boundary Conditions

Description

Insert **forces** and **fixation** to *structure*.



Step 5: Postprocessing – Directional Deformation

Description

Insert **directional deformation** to *solution* and choose **all body** as *scope geometry*.

Step 6: Postprocessing – Stress Measurement

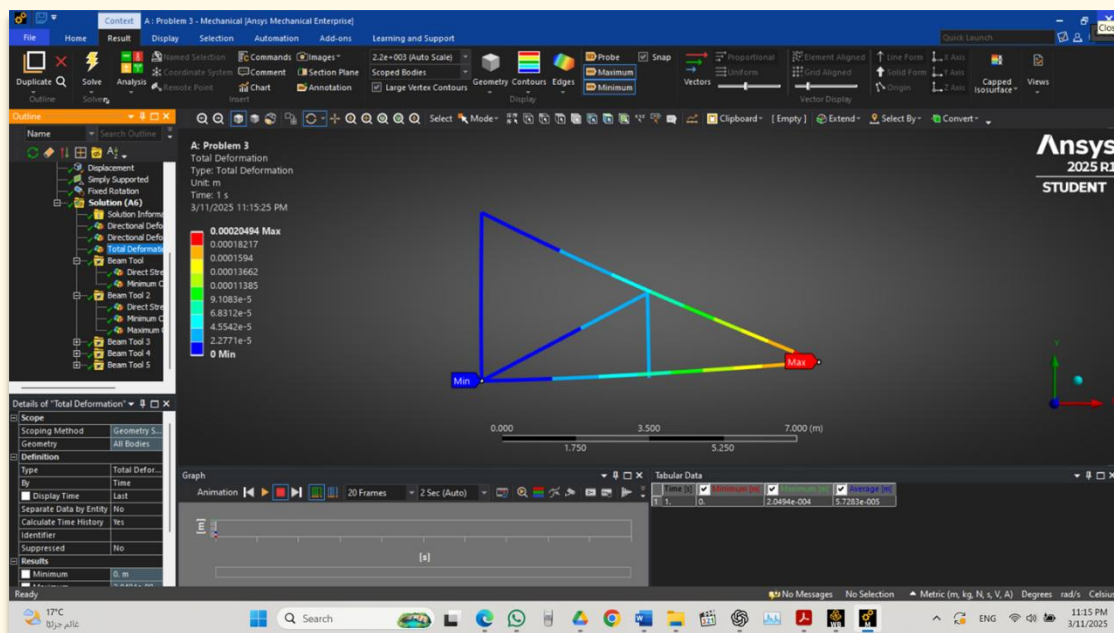
Description

Insert **beam tool** for each beam to *solution* and choose the beams of the structure as *scope geometry*.

Step 7: Postprocessing – Deformation

Description

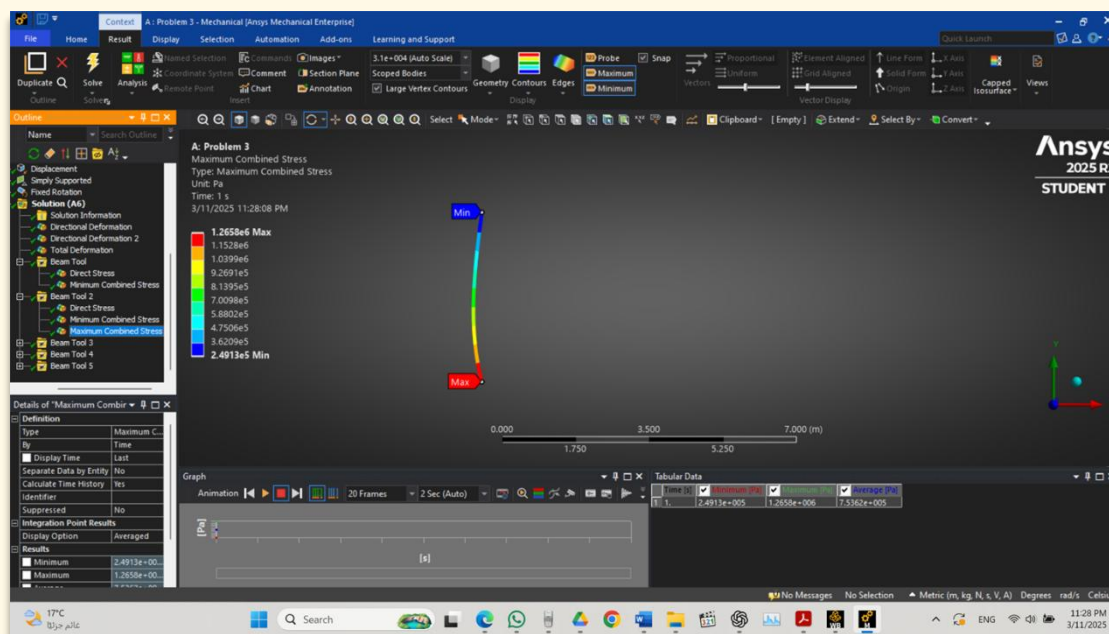
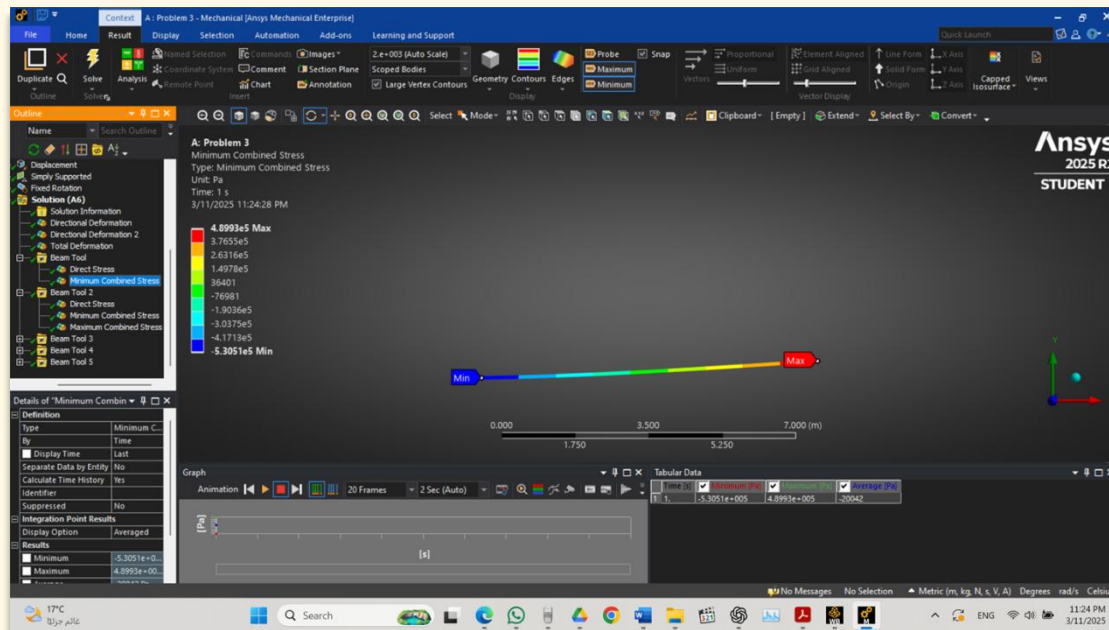
Press on solve and get total deformation.



Step 7: Postprocessing – Stresses

Description

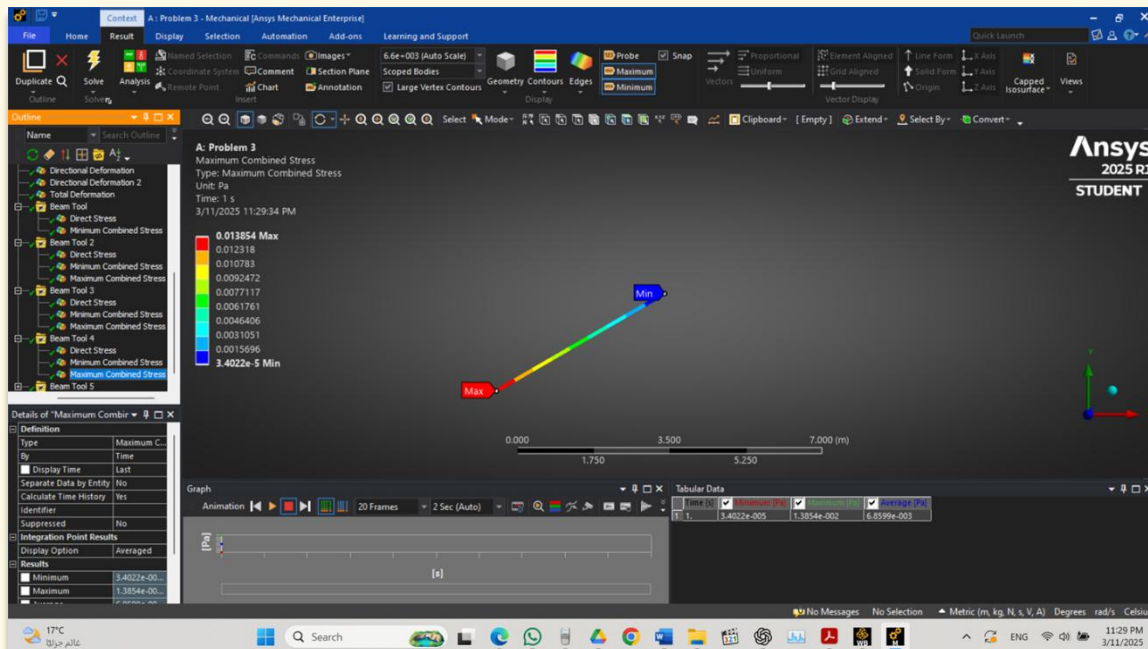
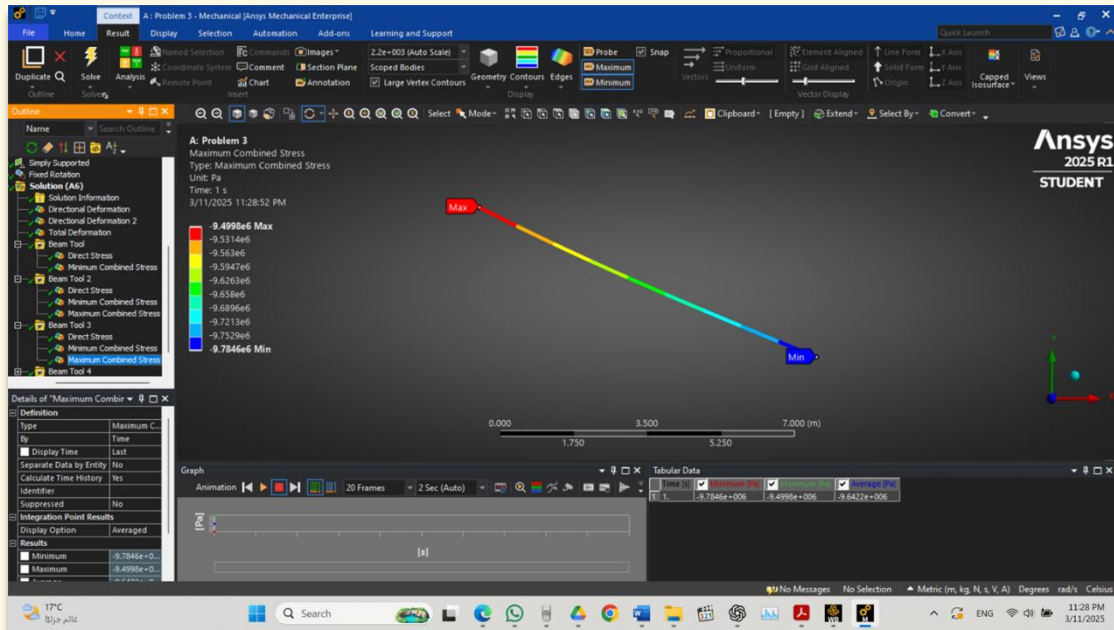
Get stress on each beam.



Step 7: Postprocessing – Stresses

Description

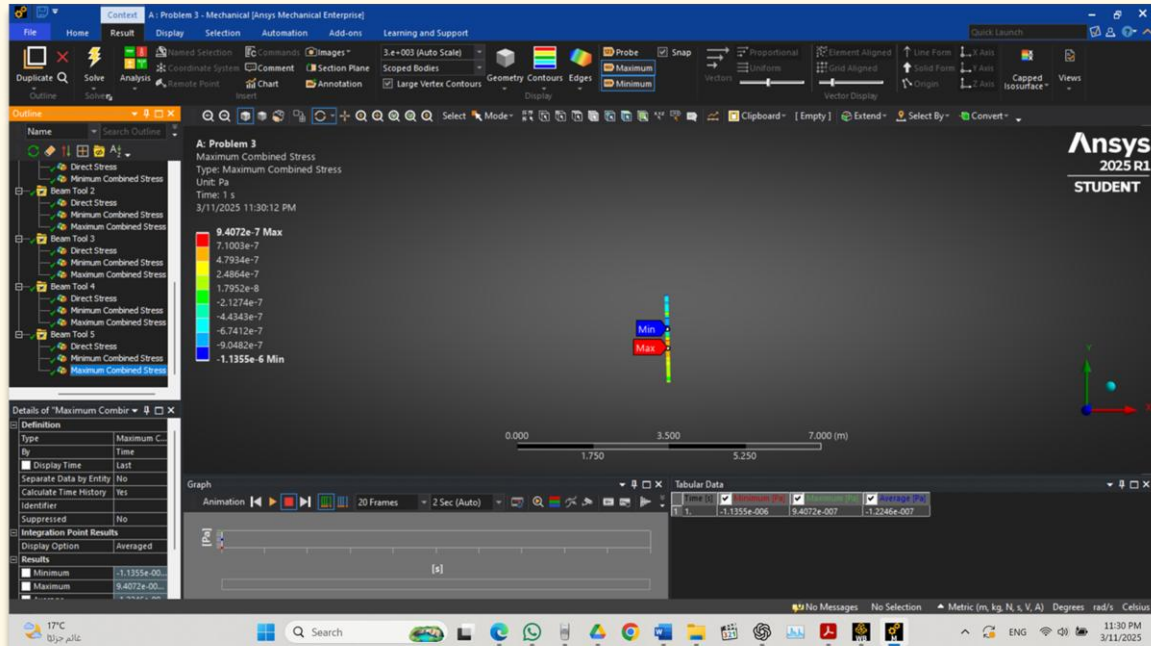
Get stress on each beam.



Step 7: Postprocessing – Stresses

Description

Get stress on each beam.



Step 8: Postprocessing – Tabulate Results

Description

Put results in a table.

Results

Total Deformation	[0.00mm, -0.21mm]
Stress on Bar 1 (1-3)	[-0.53MPa, 0.49 MPa]
Stress on Bar 2 (1-4)	[0.29 MPa, 1.27 MPa]
Stress on Bar 3 (3-4)	[-9.78 MPa, -9.50 MPa]
Stress on Bar 4 (1-5)	[0 MPa, 0 MPa]
Stress on Bar 5 (2-5)	[0 MPa, 0 MPa]

Using APDL

APDL SCRIPT Steps

Description

- Preprocessing
 - Engineering Data: adjusting real constants and material properties
 - Model: adding nodal locations, element connections, nodal constraints, and loads
- Postprocessing
 - Plotting model
 - Listing nodal displacements, element stresses, element forces.


```
/filnam, HW1_Truss
/title, Truss
/prep7
!elements
et,1,link1      ! Element type no. 1 is LINK1
!real constants
r,1,78.54      ! Real constant number 1 (cross sectional area)
!material properties
mp,ex,1,1e6    ! Material Properties: E for material no. 1
!nodal locations
n,1,0,0,0      ! Nodal location 1
n,2,0,4,0      ! Nodal location 2
n,3,4,2,0      ! Nodal location 3
n,4,4,0,0
n,5,8,0,0
!element connections
Real,1
en,1,1,2      ! Element 1 connects nodes 1 & 2
en,2,2,3
en,3,3,1
en,4,1,4
en,5,4,5
en,6,3,5
en,7,3,4
!nodal constraints
d,1,ux,0      ! node 1 allowed no displacement in x-dir
d,1,uy,0      ! node 1 allowed no displacement in y-dir
d,3,ux,0      ! node 3 allowed no displacement in x-dir
d,3,uy,0      ! node 3 allowed no displacement in y-dir
```

APDL SCRIPT

```
!loading
f,5,fy,20000      ! force on node 2 in y-dir is 20000

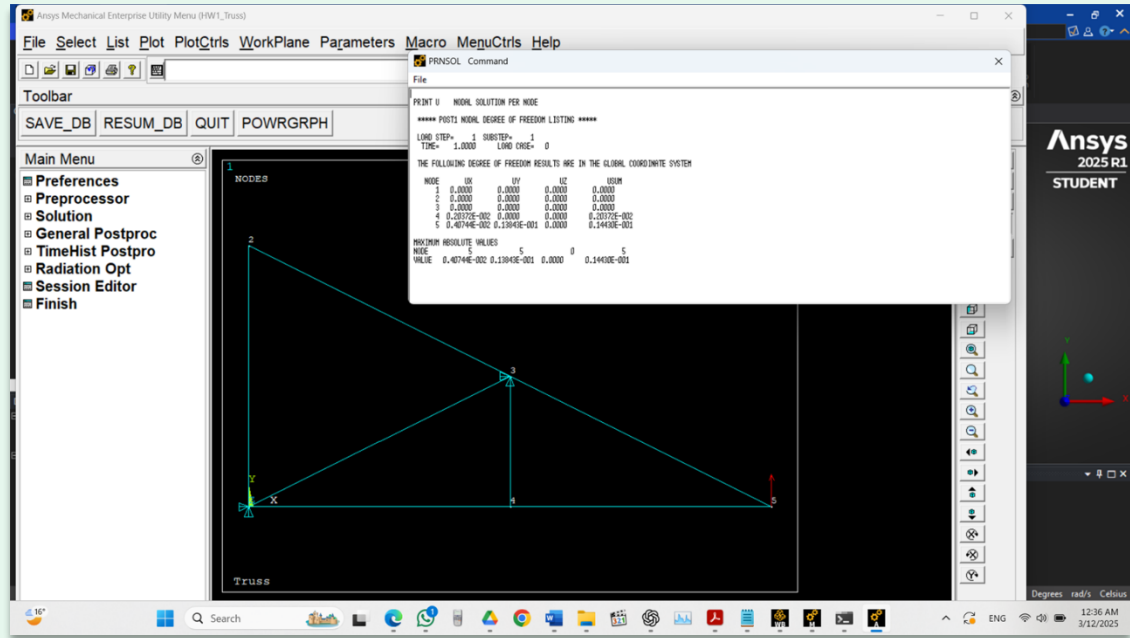
finish

/solu             ! select static load solution
antype,static
solve
save
finish

/post1           ! start the post processor

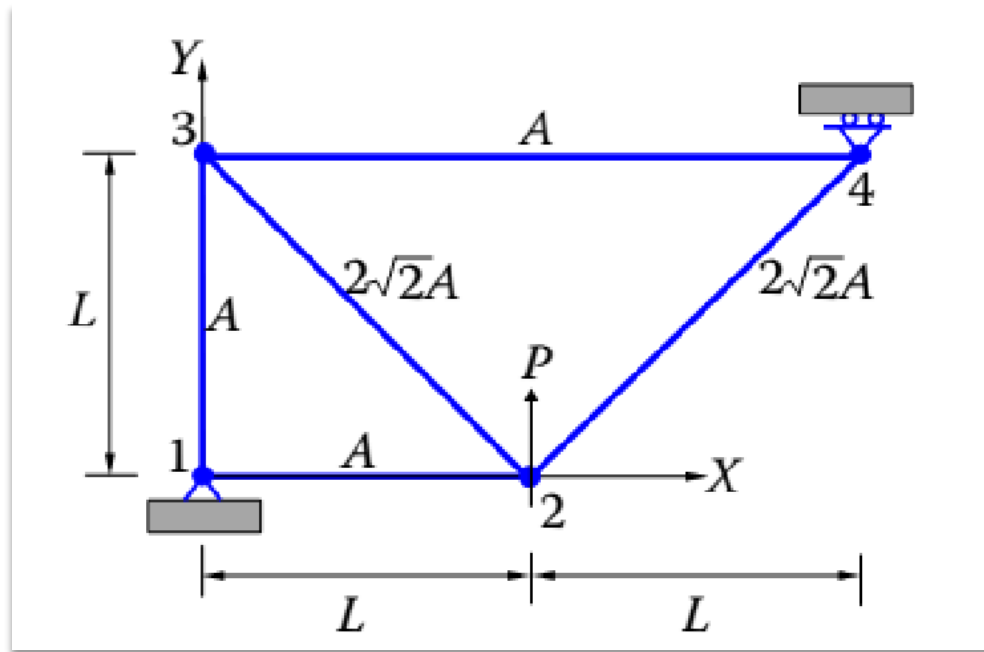
etable,Force,SMISC,1    ! create table for element axial forces
etable,Stress,LS,1      ! create table for element stress values
/output,resultsfile,dat ! define a file, resultsfile.dat, in the working
directory

prnsol,u          !write the nodal solution to file
pretab,Force,Stress
```



NODE	UX	UY	UZ	USUM
1	0	0	0	0
2	0	0	0	0
3	0	0	0	0
4	2.04E-03	0	0	2.04E-03
5	4.07E-03	1.38E-02	0	1.44E-02

Problem 4

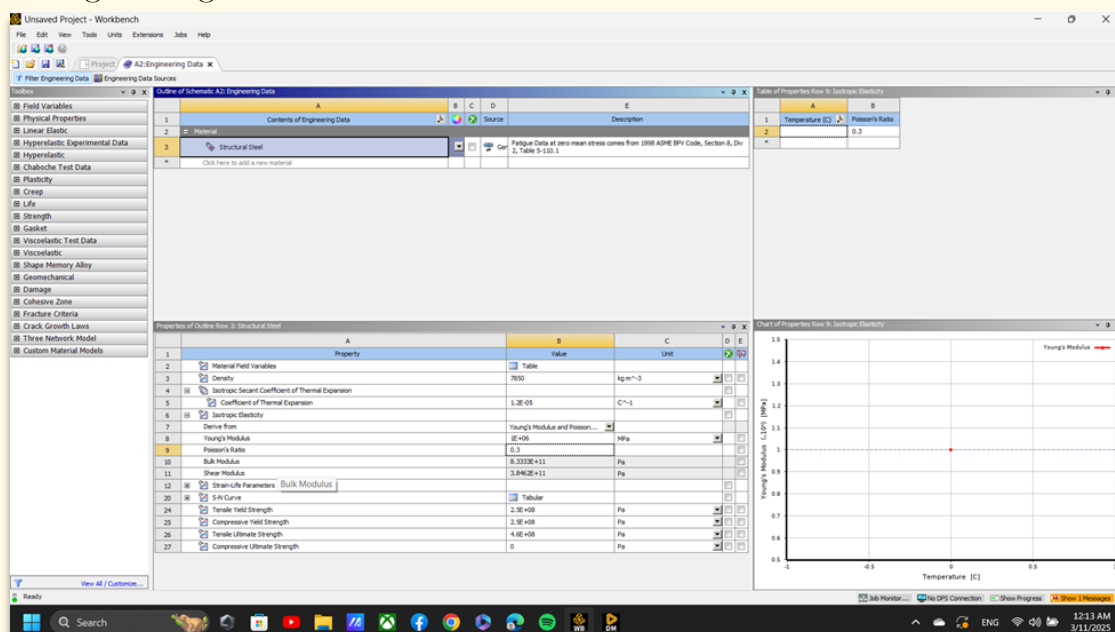


Using Workbench

Step 1: Preprocessing – Editing Engineering Data

Description

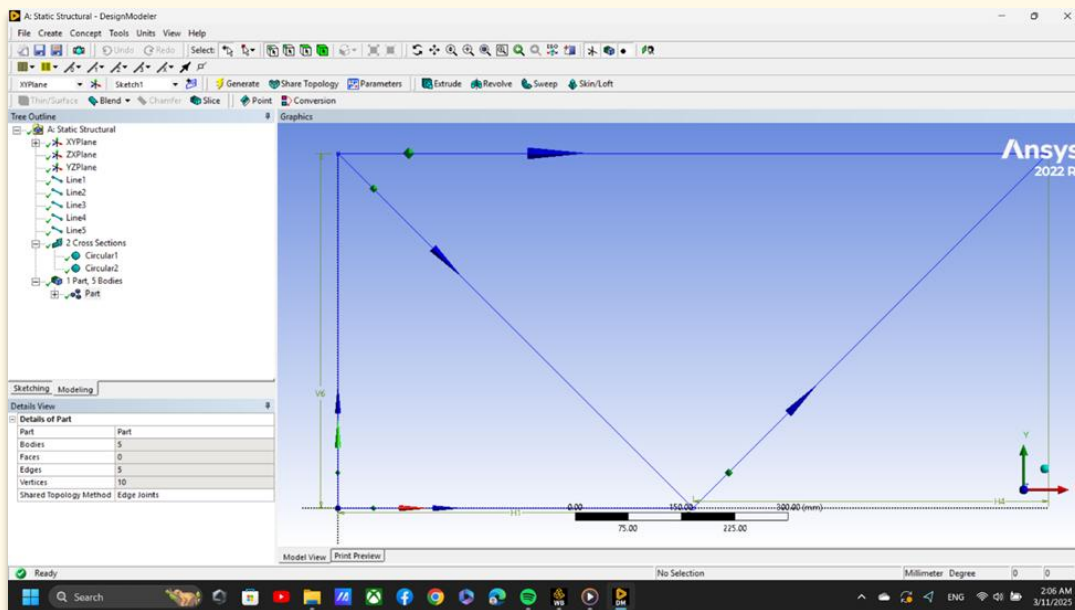
Change Young's Modulus value to 1E6 Pa.



Step 2: Preprocessing – Editing Geometry

Description

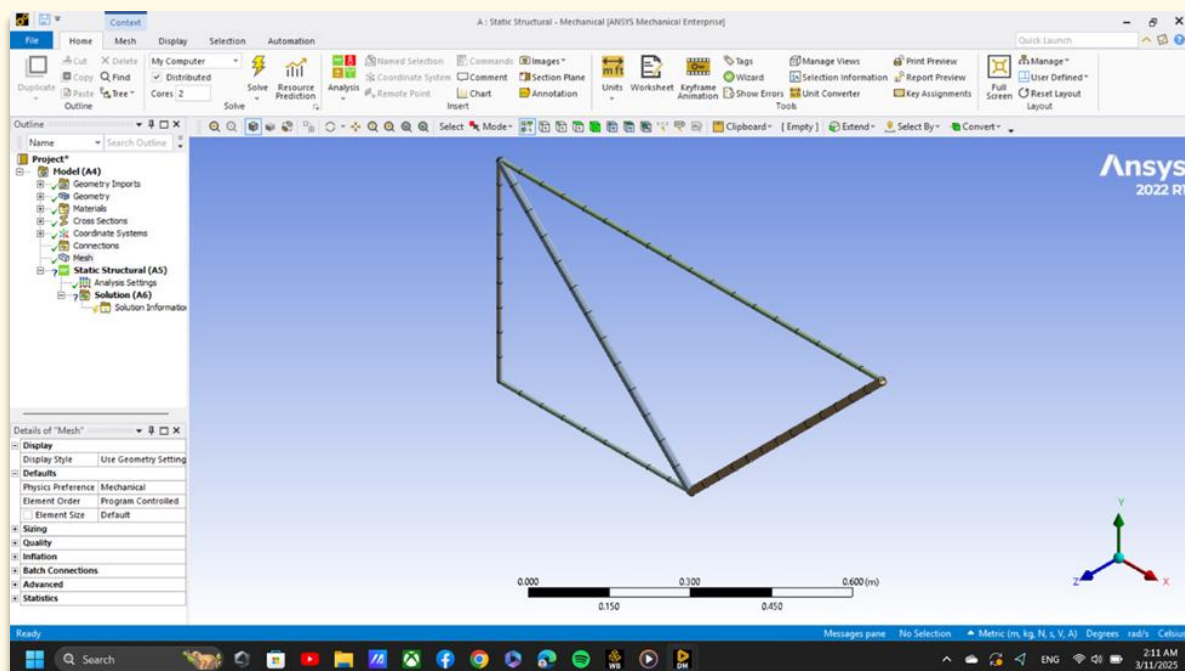
Draw structure using design modeler and add cross-section.



Step 3: Preprocessing – Editing Model – Adjusting Mesh

Description

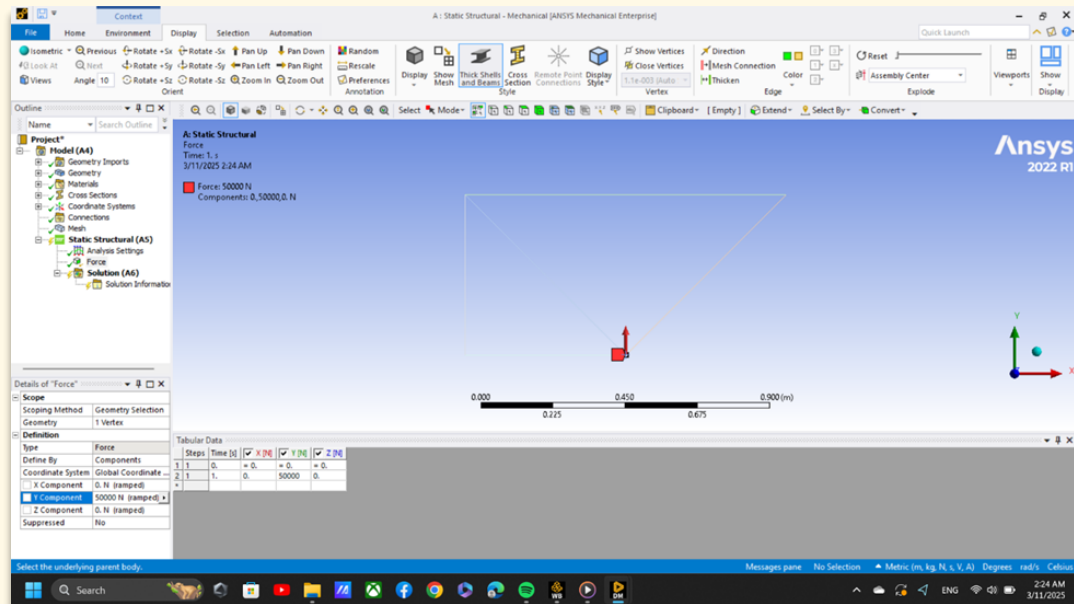
Choosing an appropriate initial **mesh resolution** for *model*, 5.



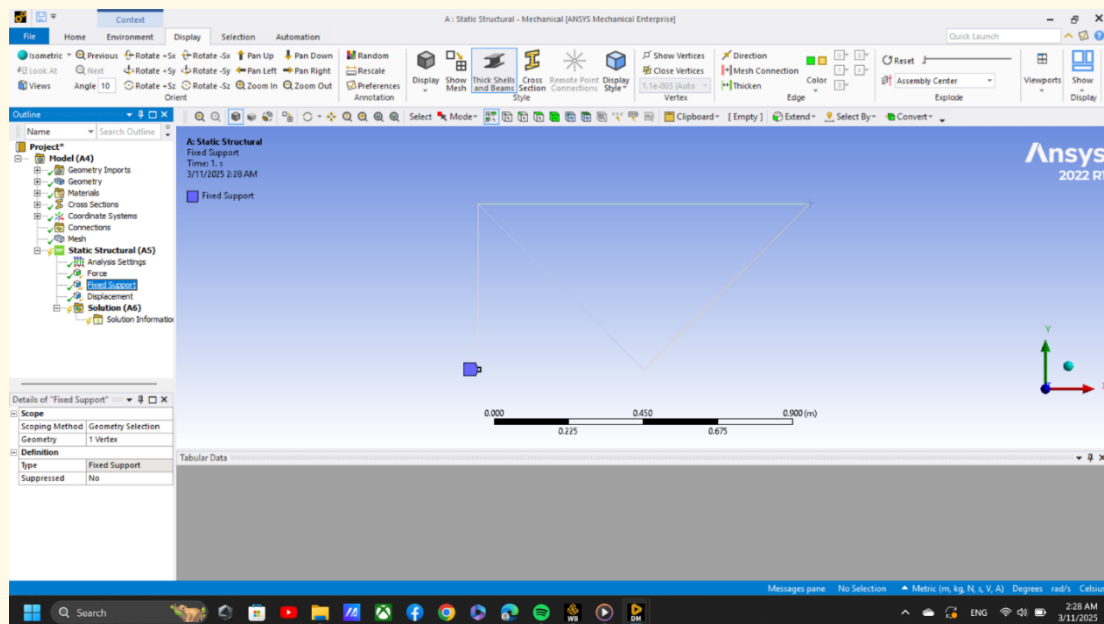
Step 4: Preprocessing – Editing Model – Boundary Conditions

Description

Insert forces and fixation to structure.



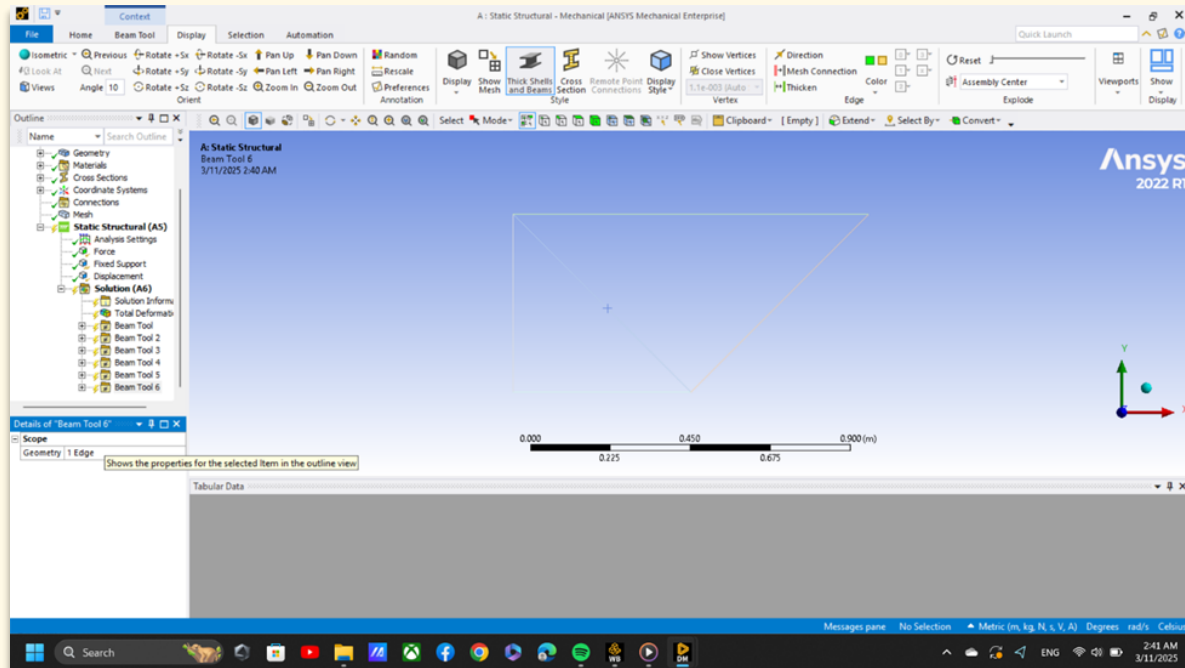
For Hinge Insert Fixed supported and for Roller Insert displacement, Set the Y and Z components to constant so that the roller doesn't allow movement there



Step 5: Postprocessing – Directional Deformation

Description

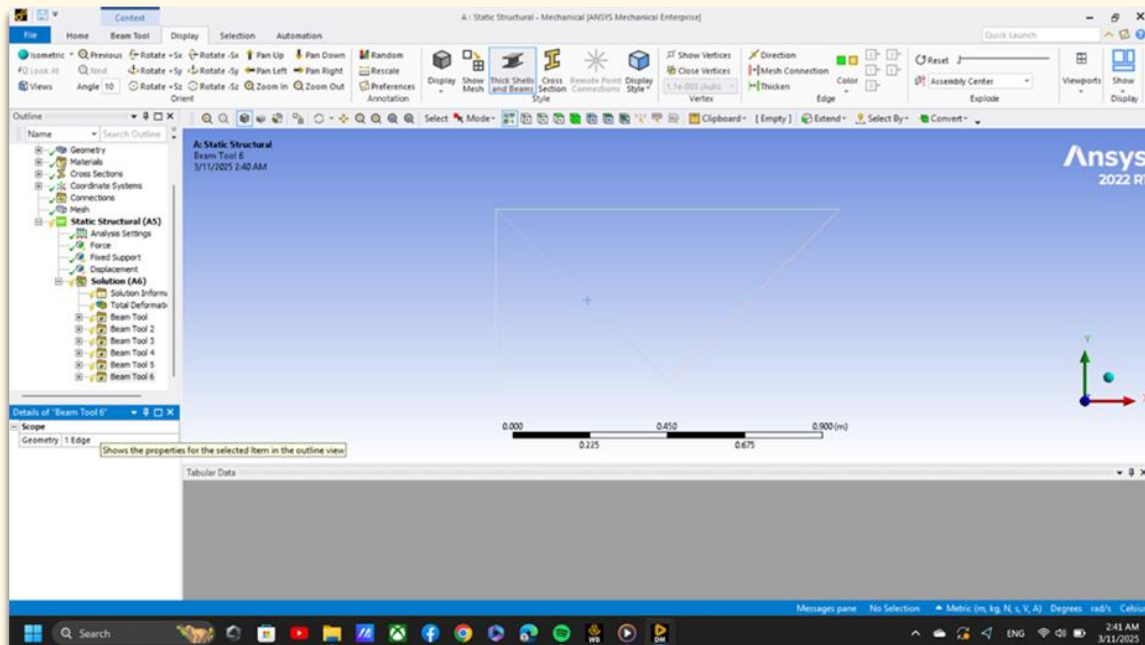
Insert **directional deformation** to *solution* and choose **all body** as *scope geometry*.



Step 6: Postprocessing – Stress Measurement

Description

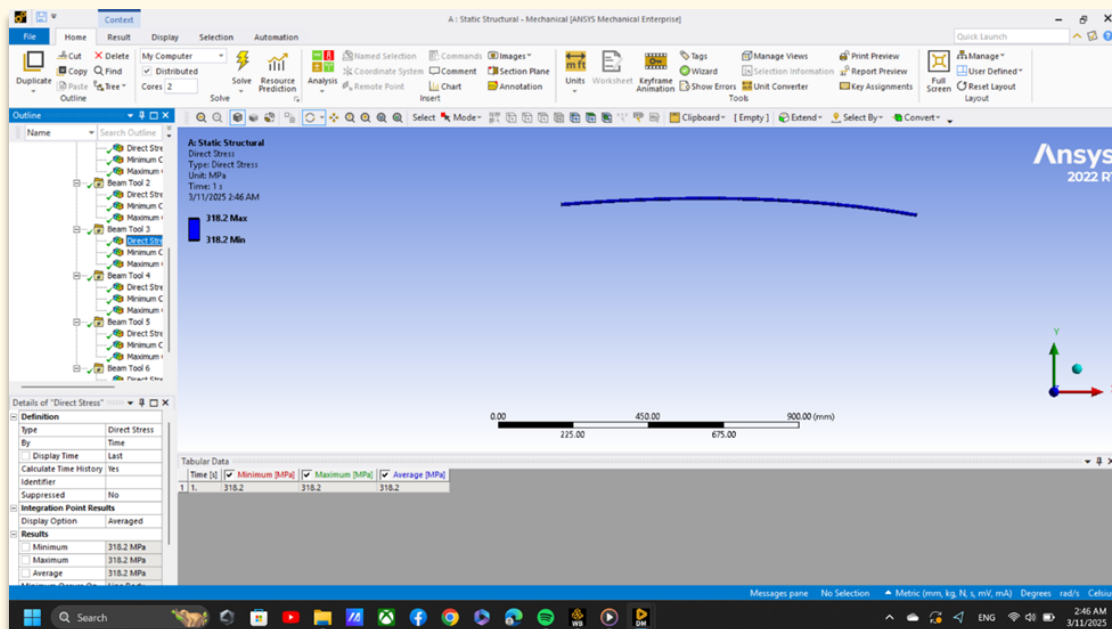
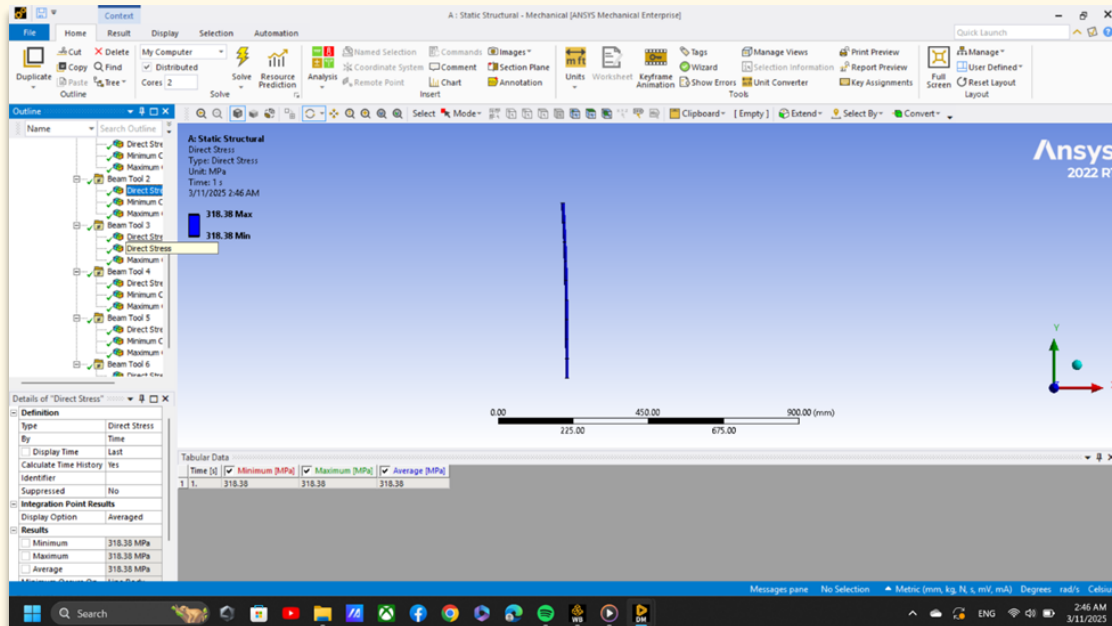
Insert **beam tool** for each beam to *solution* and choose the beams of the structure as *scope geometry*.



Step 7: Postprocessing – Stresses

Description

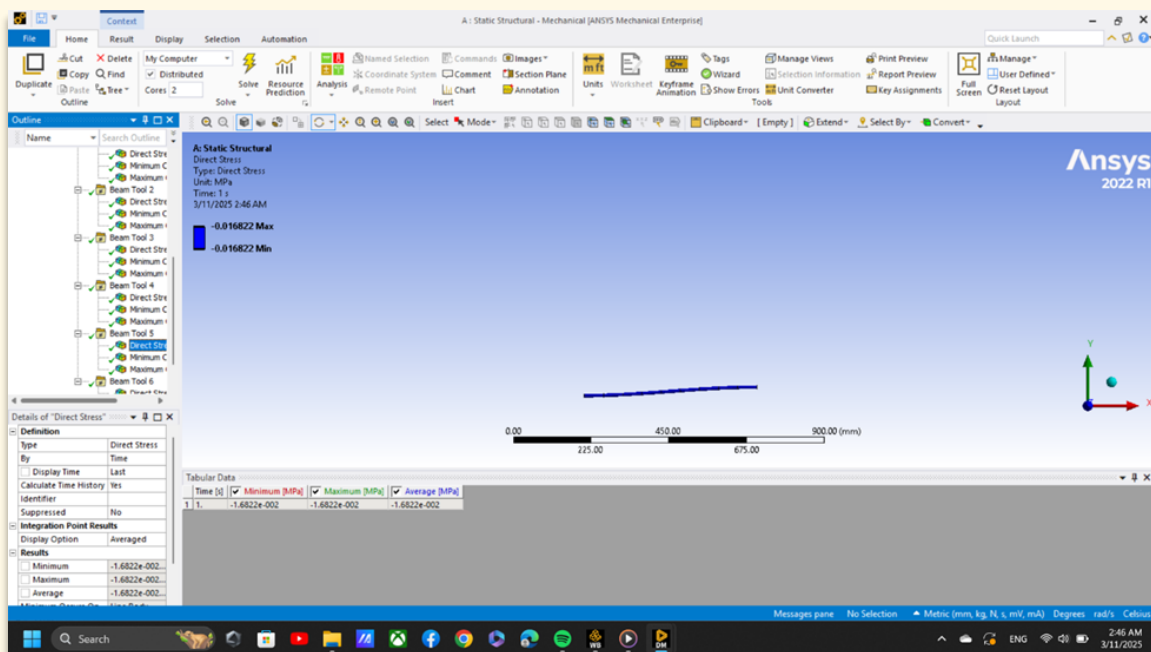
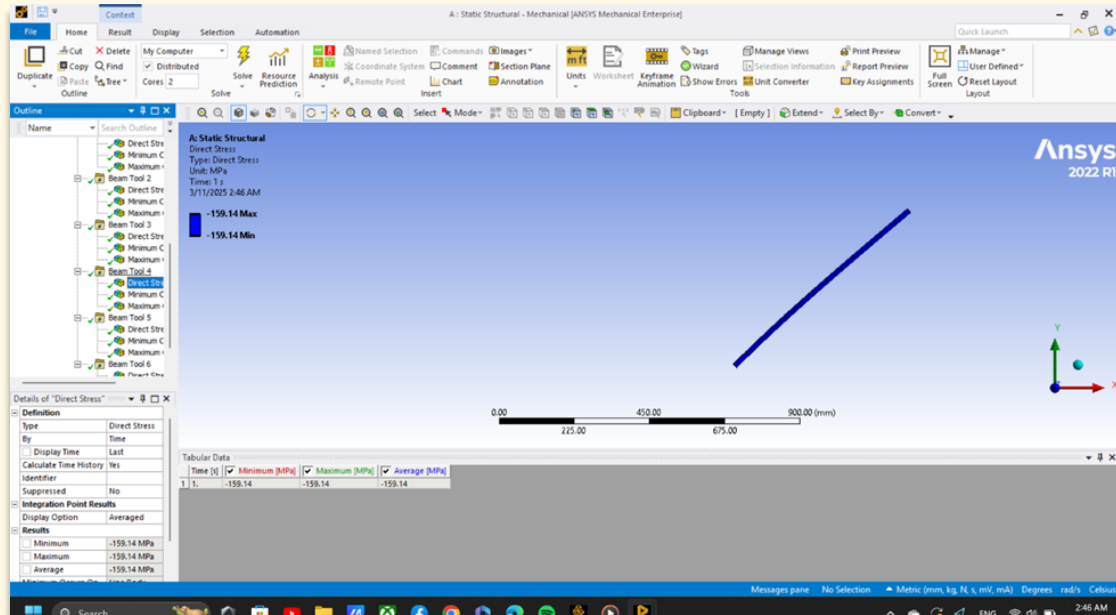
Get stress on each beam.



Step 7: Postprocessing – Stresses

Description

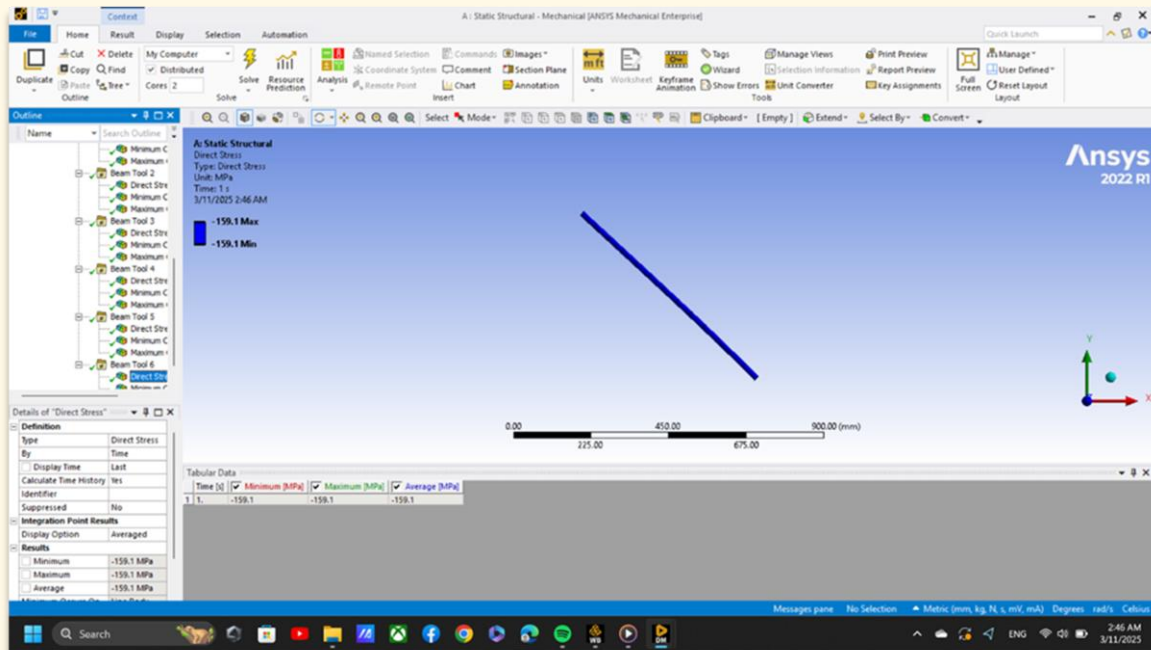
Get stress on each beam.



Step 7: Postprocessing – Stresses

Description

Get stress on each beam.



Step 8: Postprocessing – Tabulate Results

Description

Put results in a table.

Results

Total Deformation	[0.00mm, 1.55 mm]
Stress on Bar 1 (1-3)	318.38 MPa
Stress on Bar 2 (3-4)	318.2 MPa
Stress on Bar 3 (2-4)	-159.14 MPa
Stress on Bar 4 (1-2)	-0.017 MPa
Stress on Bar 5 (3-2)	159.10 MPa

Using APDL

APDL SCRIPT Steps

Description

- Preprocessing
 - Engineering Data: adjusting real constants and material properties
 - Model: adding nodal locations, element connections, nodal constraints, and loads
- Postprocessing
 - Plotting model
 - Listing nodal displacements, element stresses, element forces.

```

/filnam, HW1_Truss
/title, Pb 4
/prep7

!elements
et,1,link1      ! Element type no. 1 is LINK1

!real constants
r,1,78.54      ! Real constant number 1 (cross sectional area)

!material properties
mp,ex,1,1e6    ! Material Properties: E for material no. 1

!nodal locations
n,1,0,0,0      ! Nodal location 1
n,2,500,0,0    ! Nodal location 2
n,3,0,500,0    ! Nodal location 3
n,4,1000,500,0 ! Nodal location 4

!element connections
Real,1
en,1,1,2      ! Element 1 connects nodes 1 & 2
en,2,2,3      ! Element 2 connects nodes 2 & 3
en,3,1,3      ! Element 2 connects nodes 2 & 3
en,4,2,4      ! Element 2 connects nodes 2 & 3
en,5,3,4      ! Element 2 connects nodes 2 & 3

!nodal constraints
d,1,ux,0      ! node 1 allowed no displacement in x-dir
d,1,uy,0      ! node 1 allowed no displacement in y-dir
d,3,uy,0      ! node 3 allowed no displacement in y-dir
d,4,ux,0      ! node 4 allowed no displacement in x-dir
d,4,uy,0      ! node 4 allowed no displacement in y-dir

```

APDL SCRIPT

```
!loading
f,2,fy,50000      ! force on node 2 in y-dir is 50000

finish

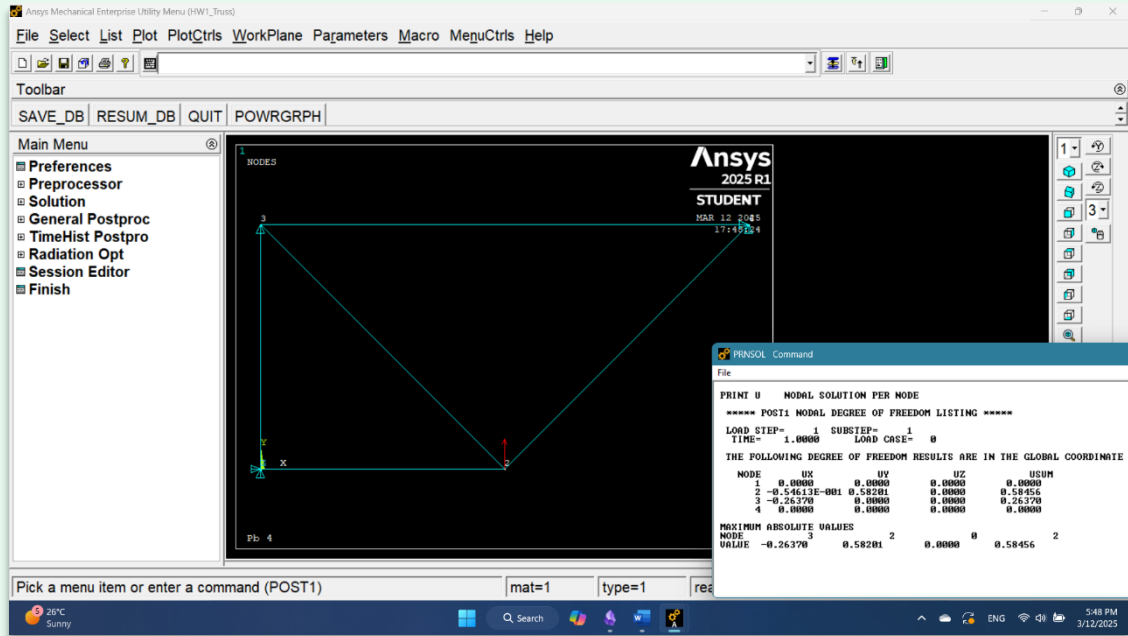
/solu             ! select static load solution
antype,static
solve
save
finish

/post1           ! start the post processor

etable,Force,SMISC,1      ! create table for element axial forces
etable,Stress,LS,1       ! create table for element stress values

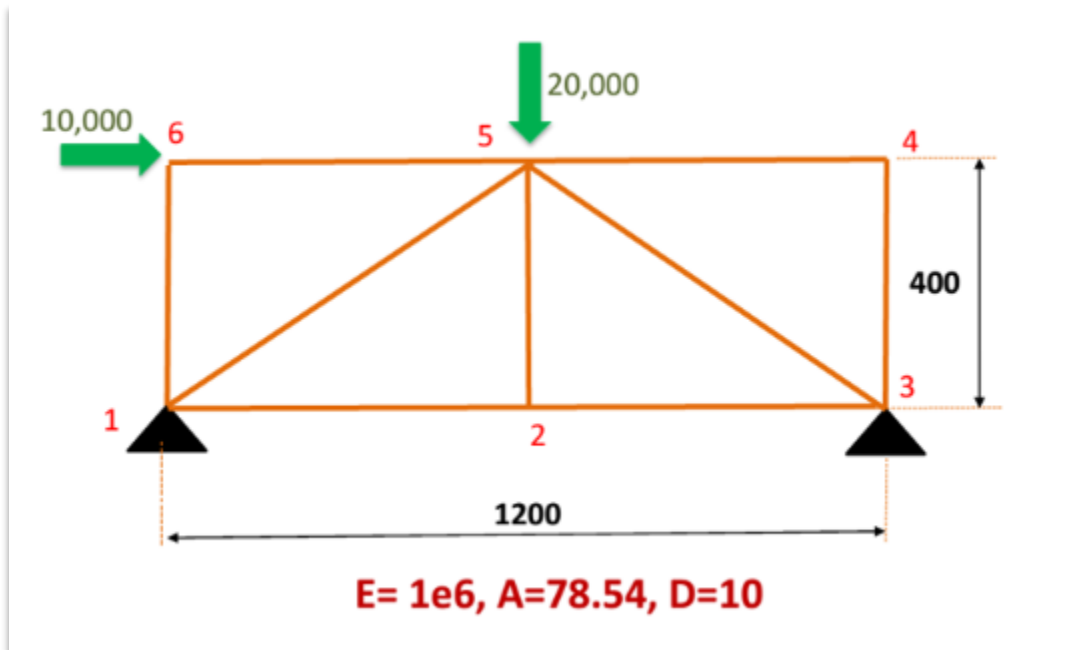
/output,resultsfile,dat  ! define a file, resultsfile.dat, in the working
directory

prnsol,u          !write the nodal solution to file
pretab,Force,Stress
```



NODE	UX	UY	UZ	USUM
1	0	0	0	0
2	-5.46E-02	0.58201	0	0.58456
3	-0.2637	0	0	0.2637
4	0	0	0	0

Problem 5

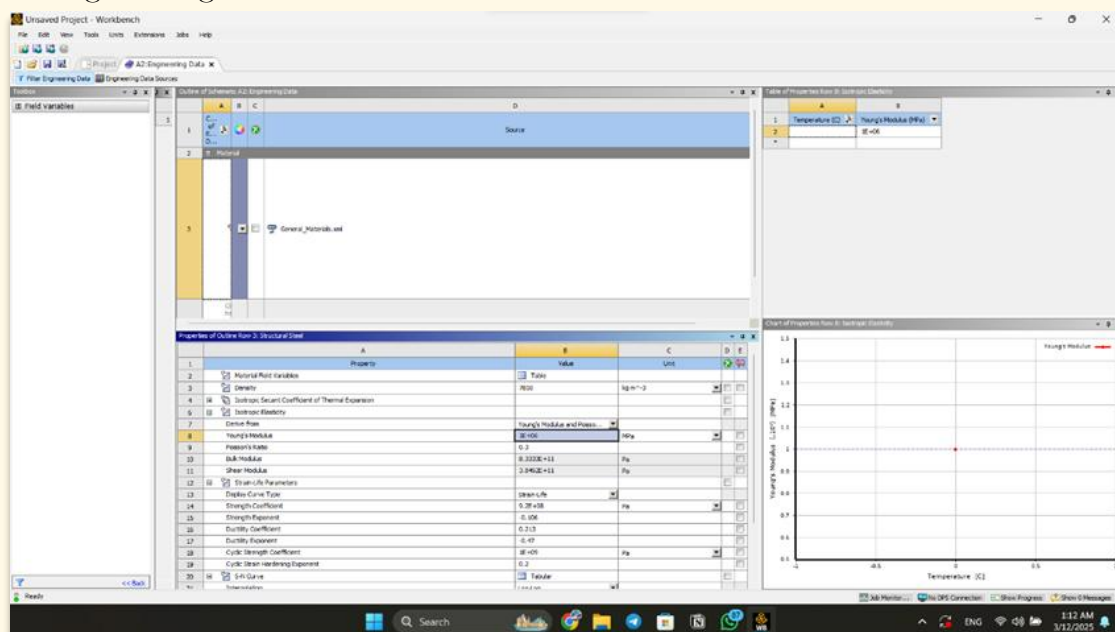


Using Workbench

Step 1: Preprocessing – Editing Engineering Data

Description

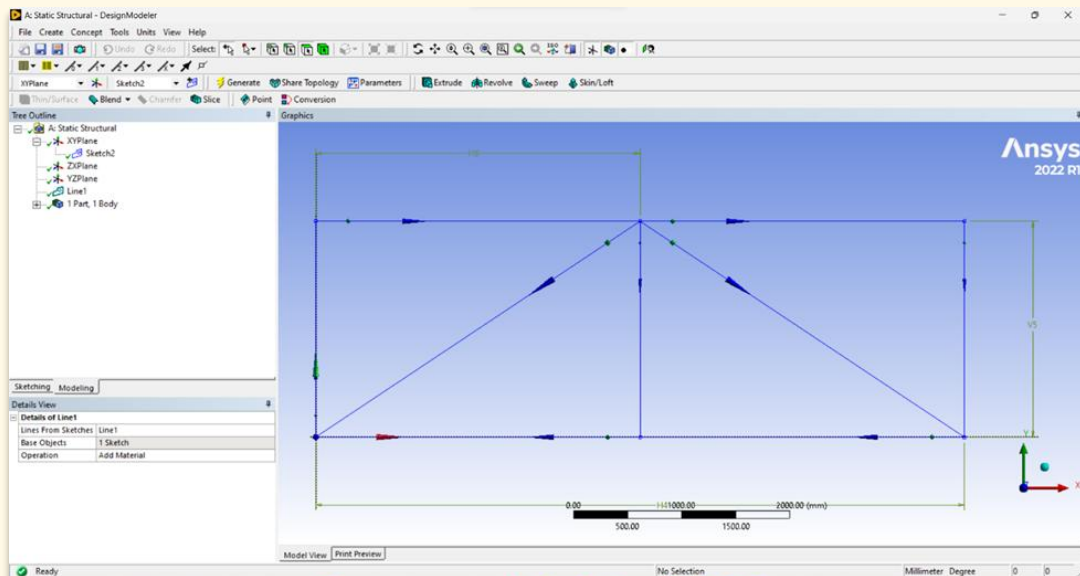
Change Young's Modulus value to $1E6$ Pa.



Step 2: Preprocessing – Editing Geometry

Description

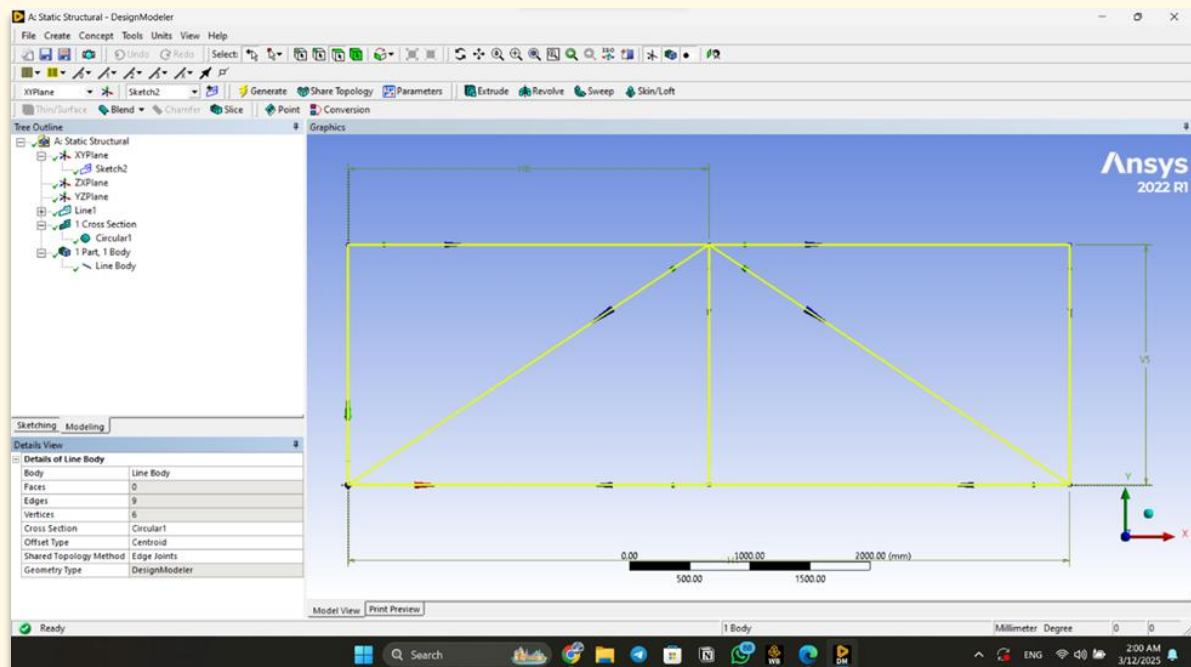
Draw structure using design modeler and add cross-section.



Step 3: Preprocessing – Editing Model – Adjusting Mesh

Description

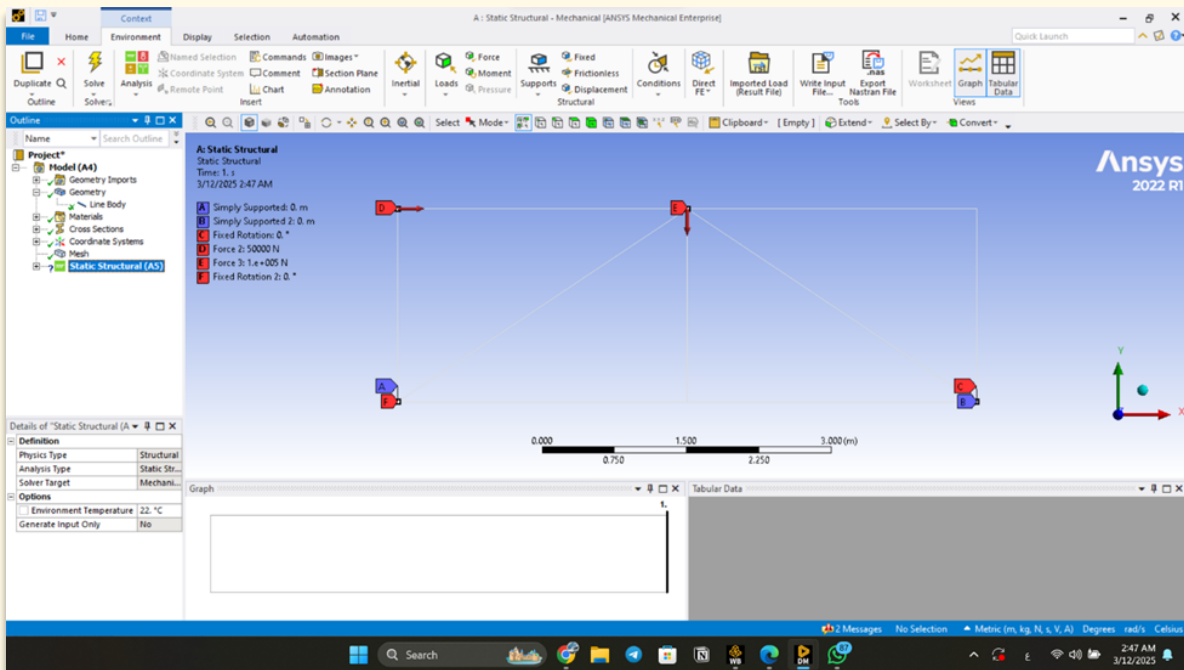
Choosing an appropriate initial **mesh resolution** for *model*, 5.



Step 4: Preprocessing – Editing Model – Boundary Conditions

Description

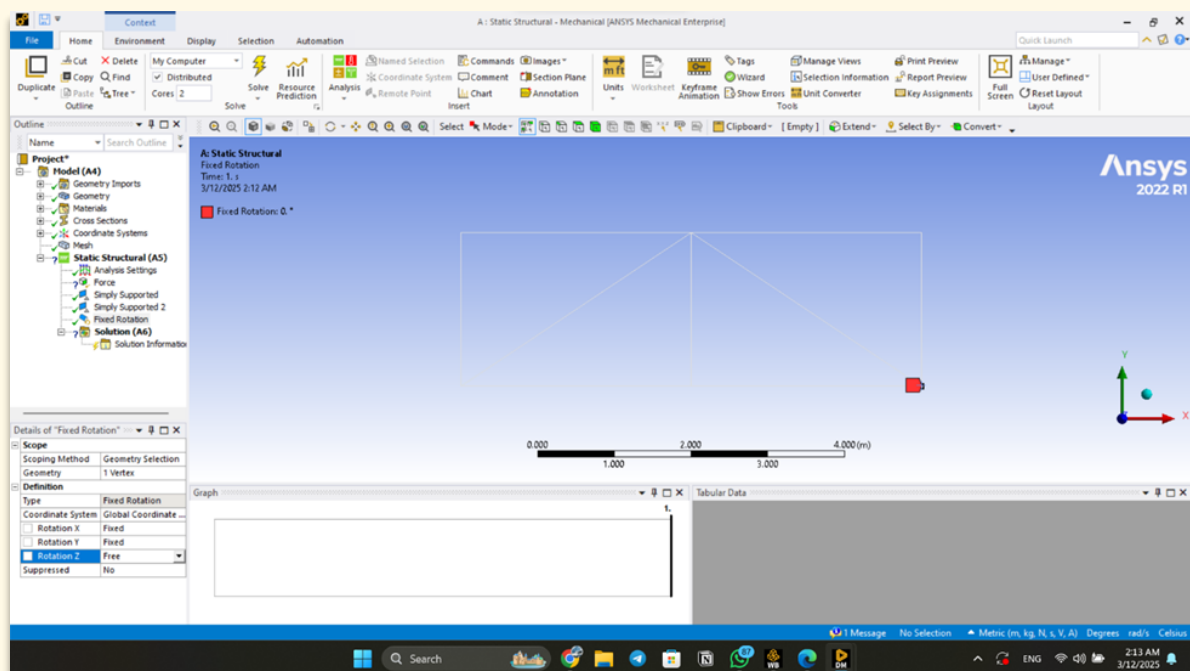
Insert **forces** and **fixation** to *structure*.



Step 5: Postprocessing – Directional Deformation

Description

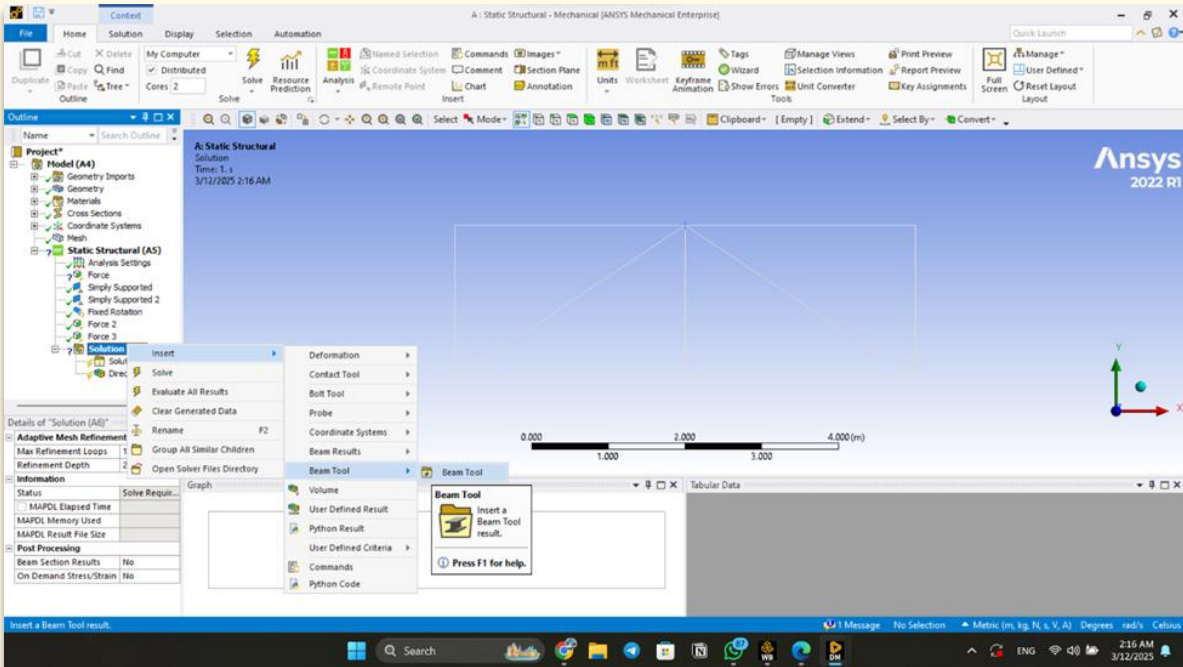
Insert **directional deformation** to *solution* and choose **all body** as *scope geometry*.



Step 6: Postprocessing – Stress Measurement

Description

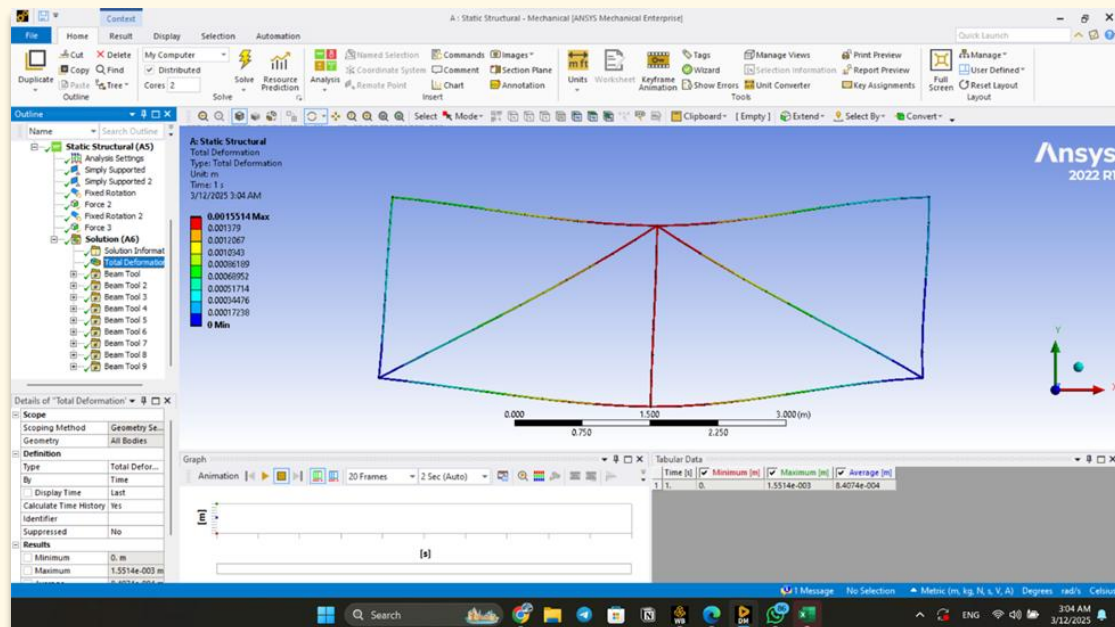
Insert **beam tool** for each beam to *solution* and choose the beams of the structure as *scope geometry*.



Step 7: Postprocessing – Deformation

Description

Press on solve and get total deformation.

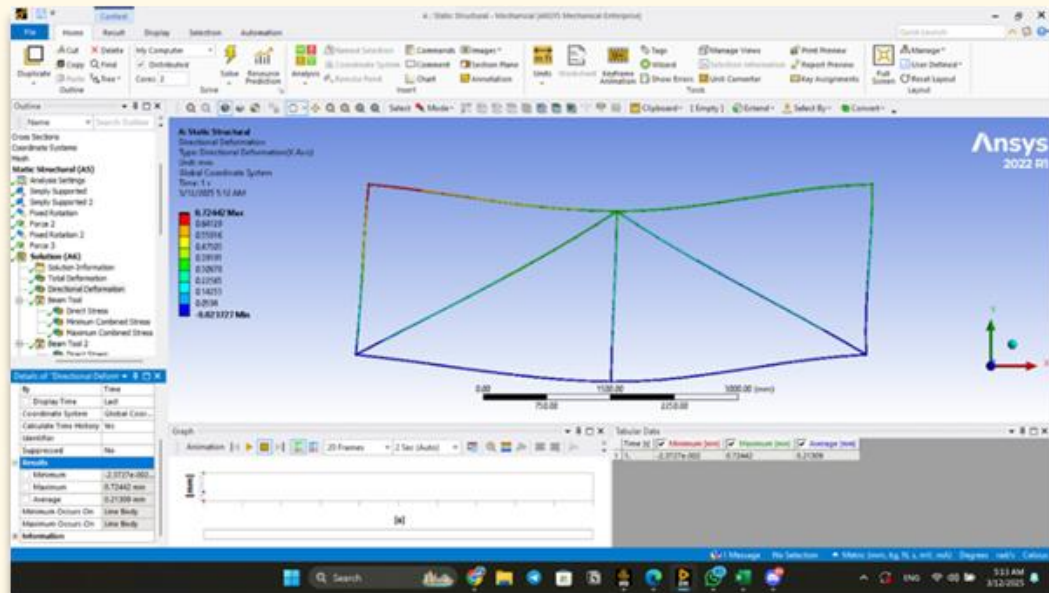


Step 7: Postprocessing – Deformation

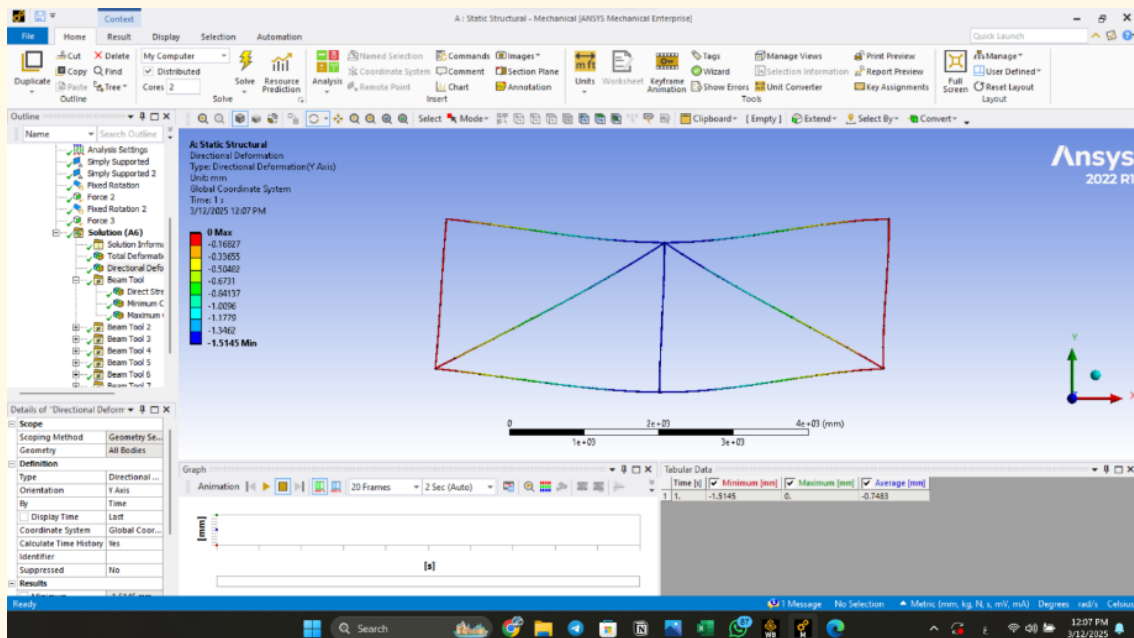
Description

Get directional deformation.

x-axis



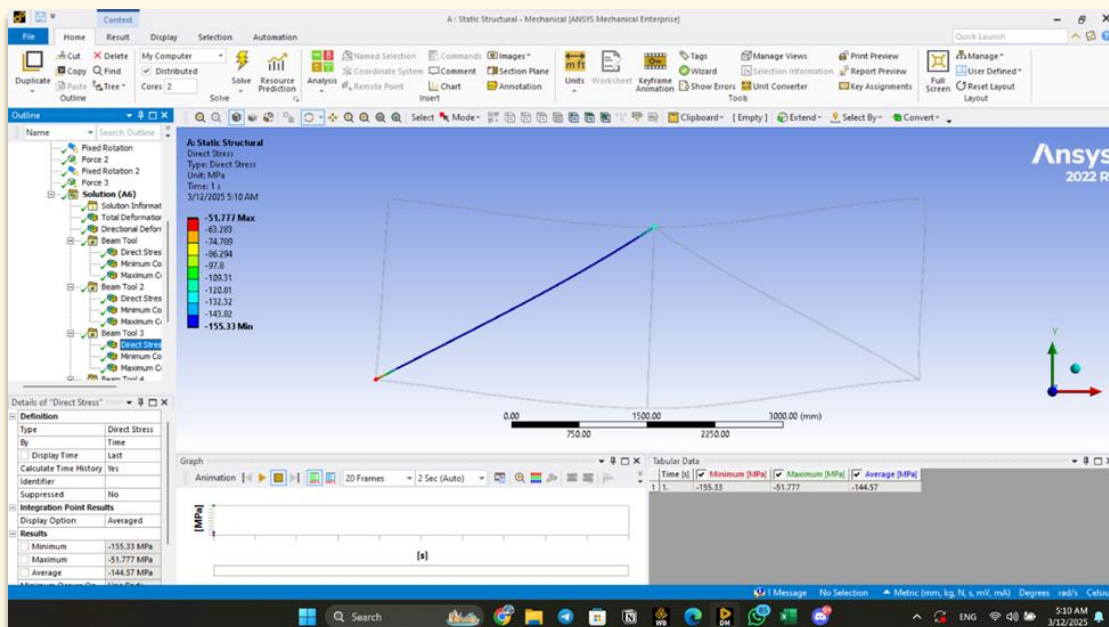
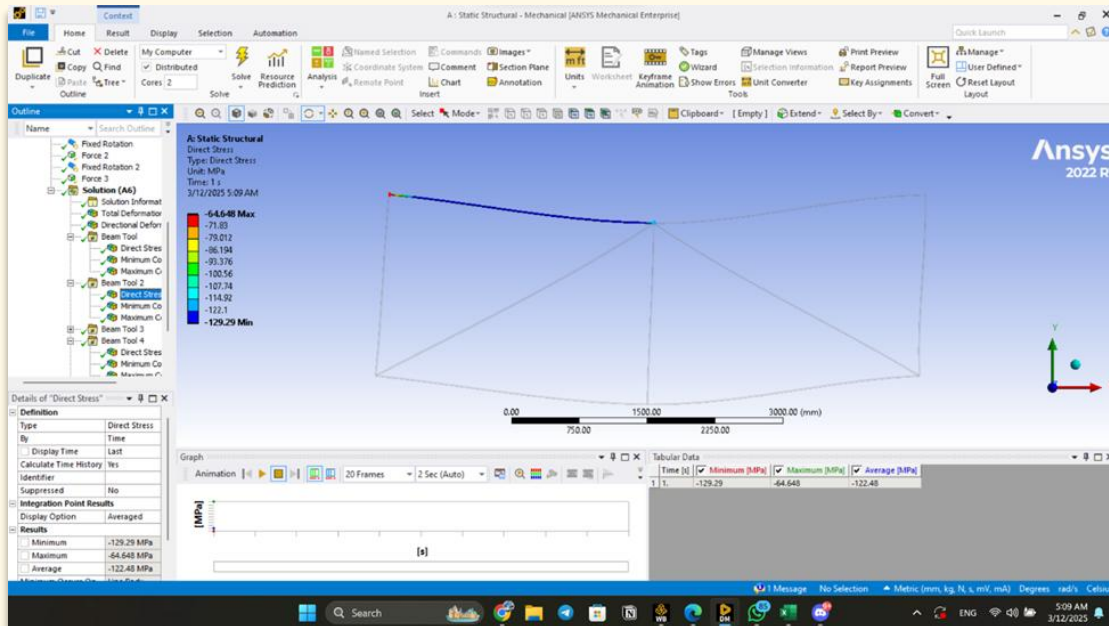
y-axis



Step 7: Postprocessing – Stresses

Description

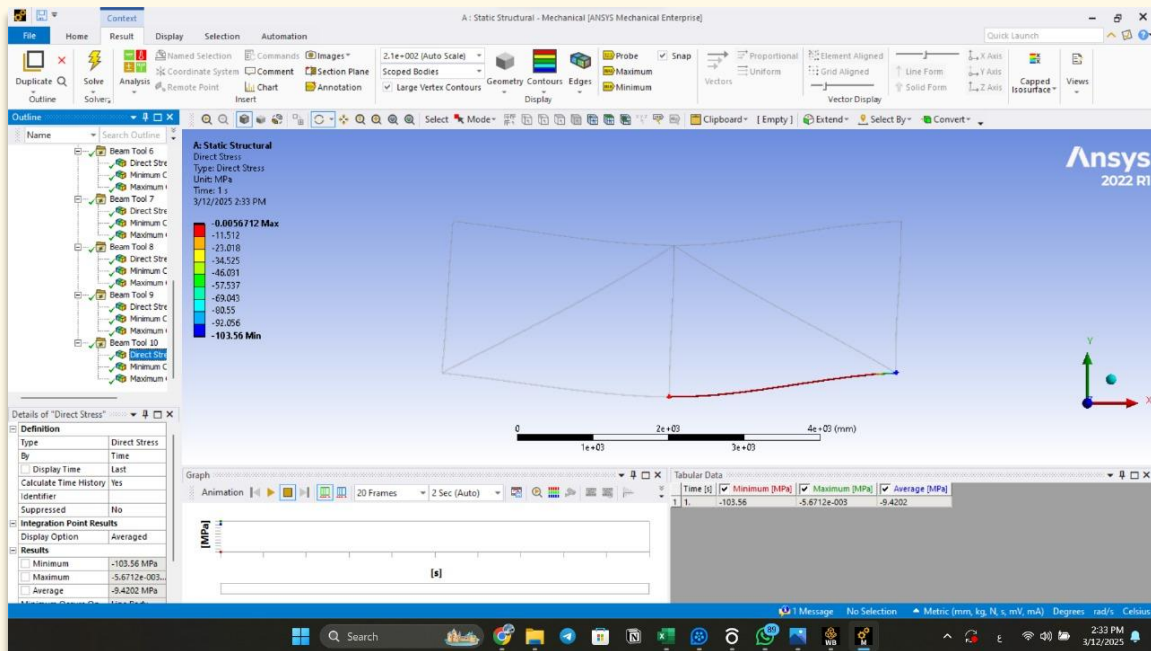
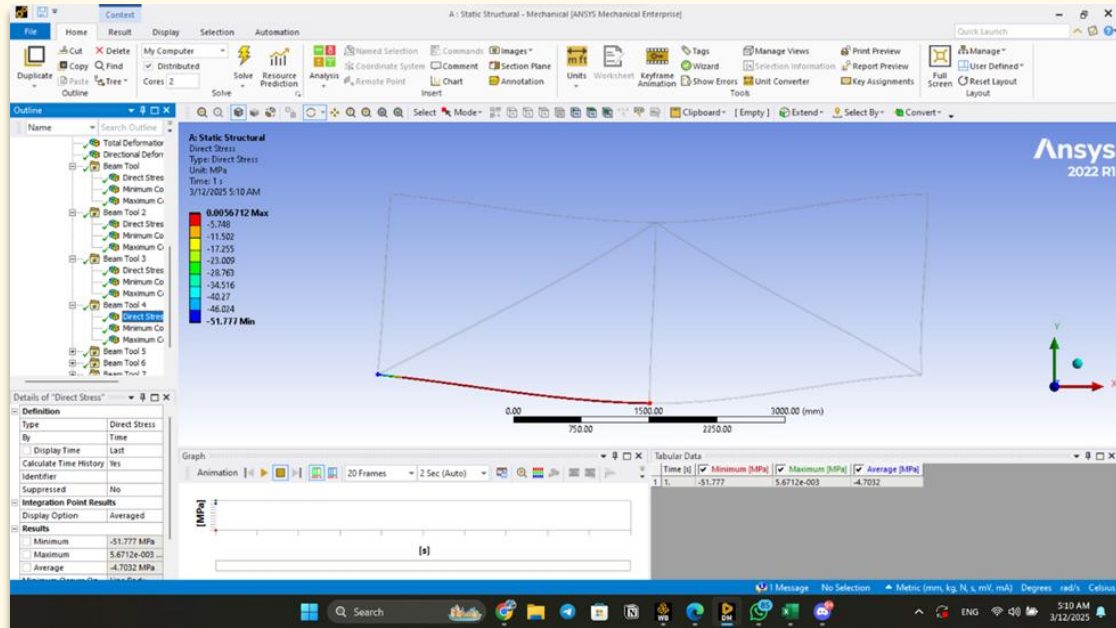
Get stress on each beam.



Step 7: Postprocessing – Stresses

Description

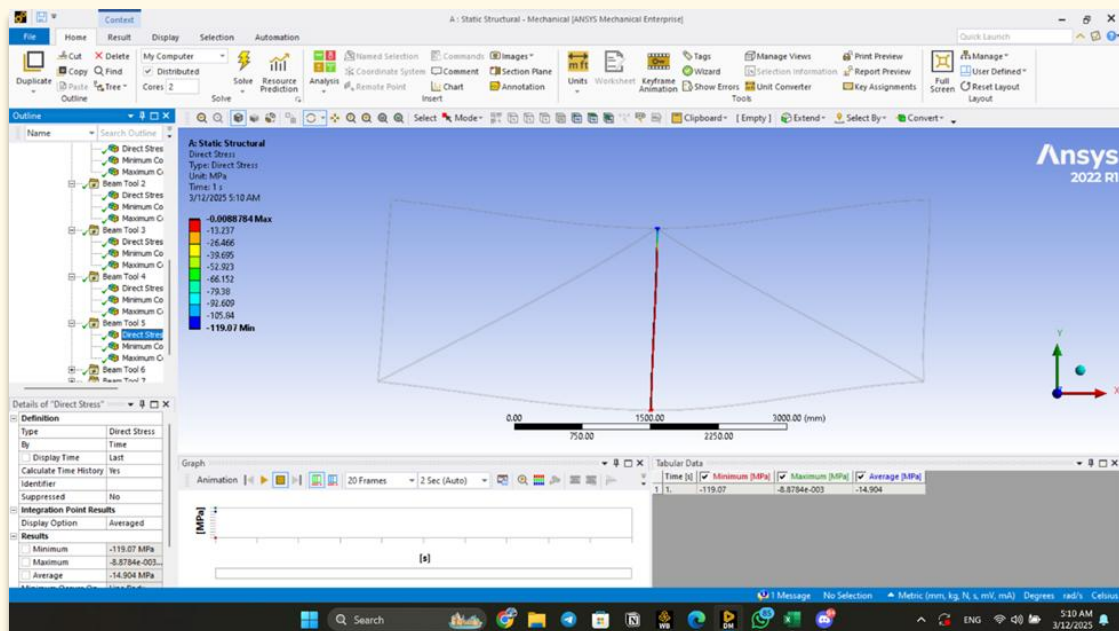
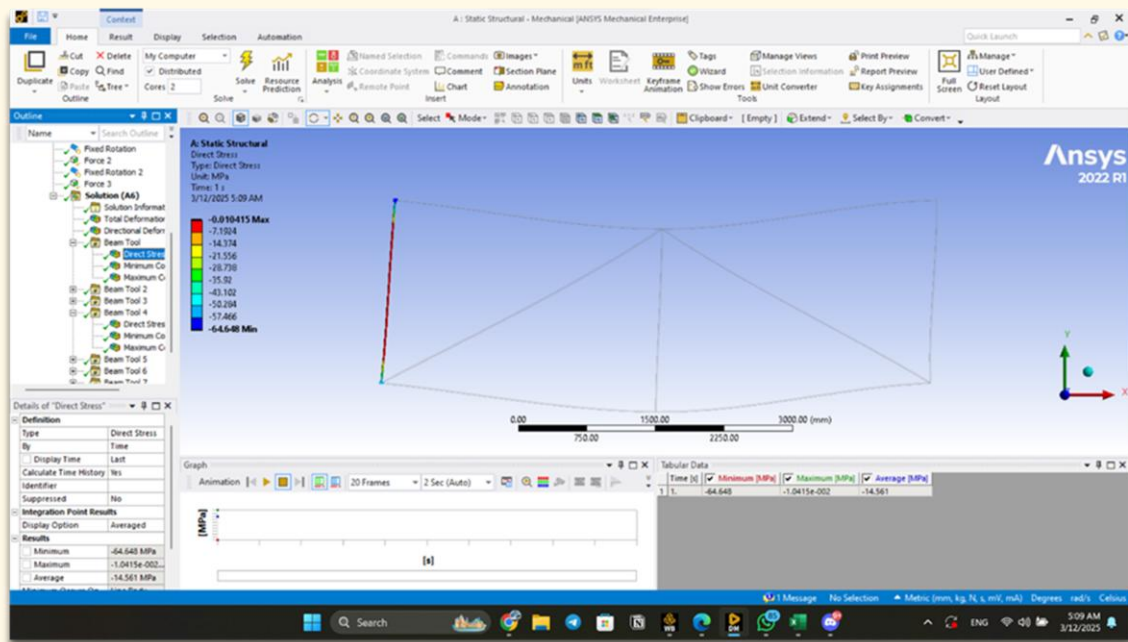
Get stress on each beam.



Step 7: Postprocessing – Stresses

Description

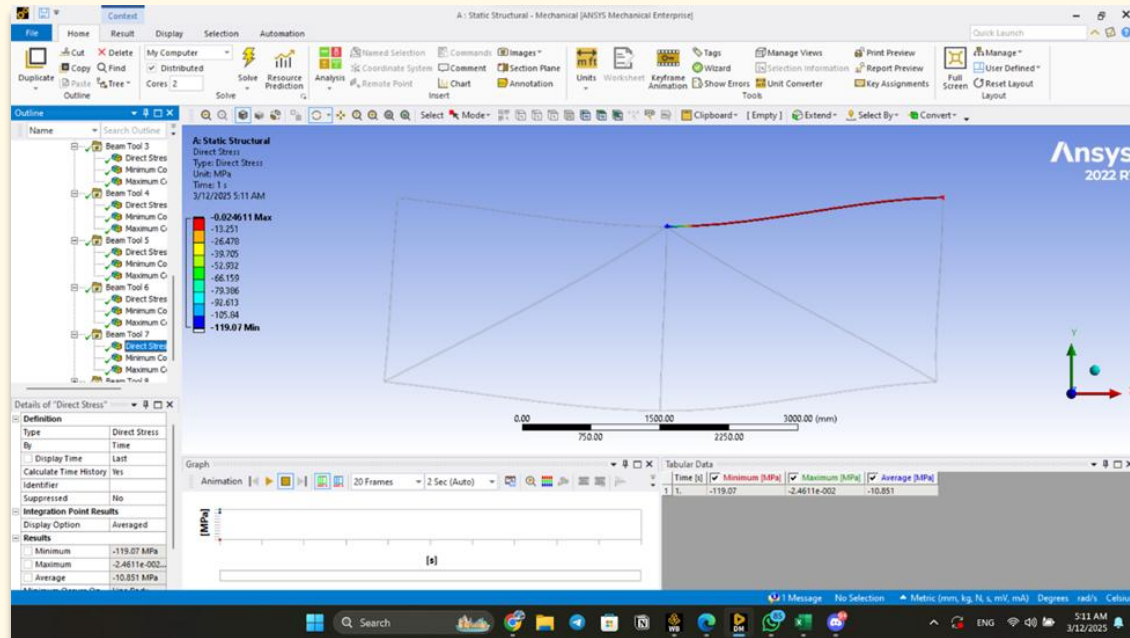
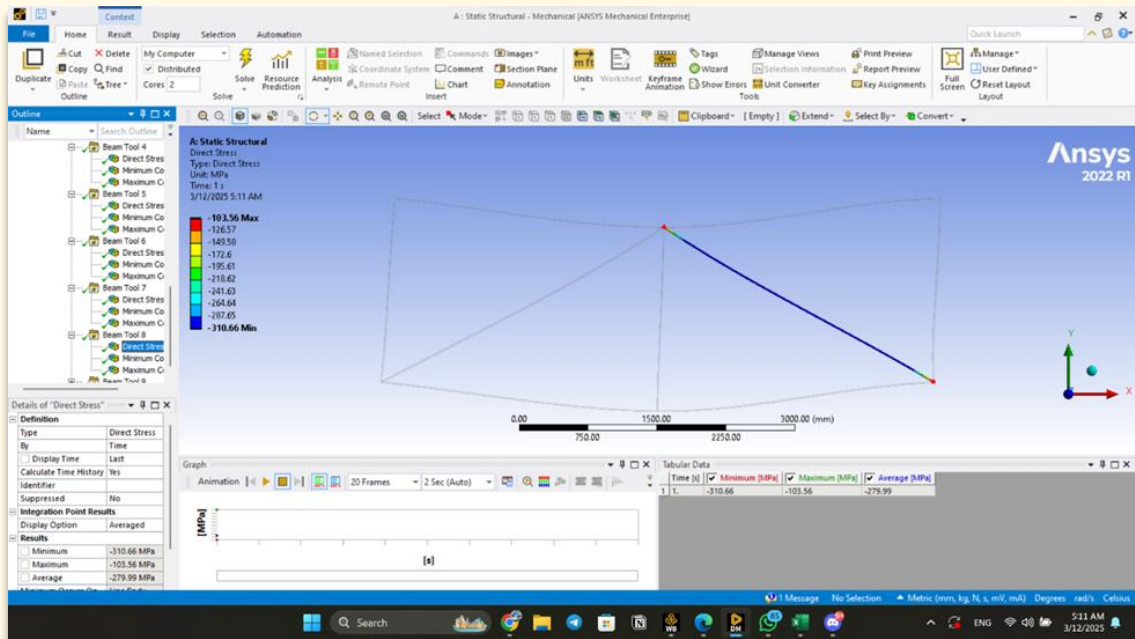
Get stress on each beam.



Step 7: Postprocessing – Stresses

Description

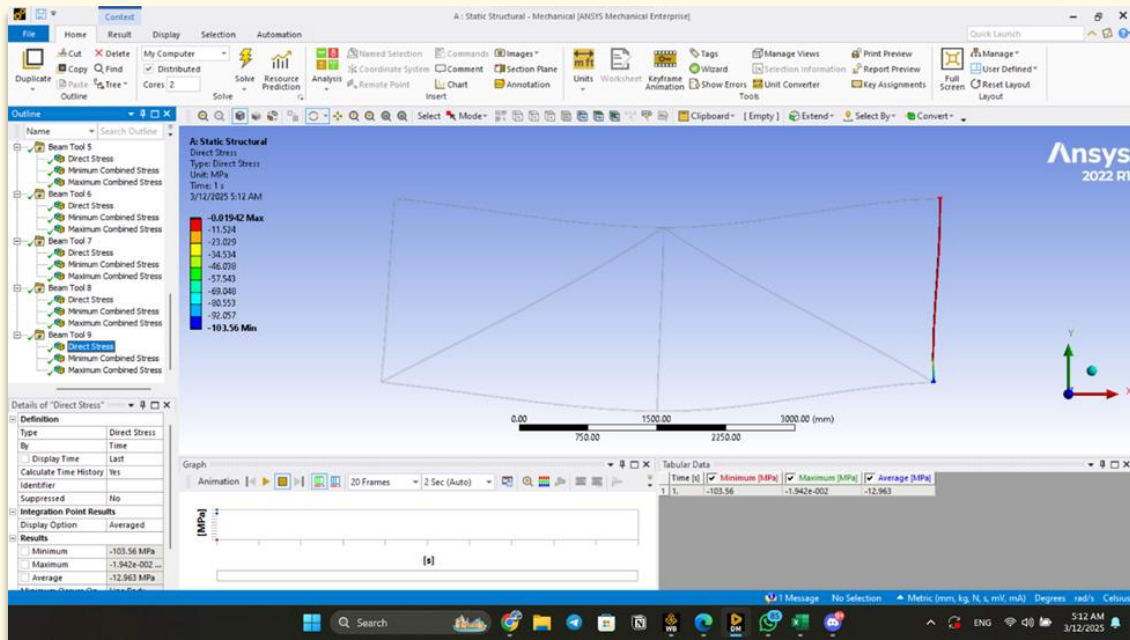
Get stress on each beam.



Step 7: Postprocessing – Stresses

Description

Get stress on each beam.



Step 8: Postprocessing – Tabulate Results

Description

Put results in a table.

Results

Directional Deformation (y)	[-1.51mm, -0.17mm]
Directional Deformation (x)	[-0.023mm, 0.724 mm]
Total Deformation	[0.00mm, 1.55 mm]
Stress on Bar 1 (1-2)	[-51.78 MPa, 0.0057 MPa]
Stress on Bar 2 (2-3)	[-103.56 MPa, -0.0057 MPa]
Stress on Bar 3 (3-4)	[-103.56 MPa, -0.019 MPa]
Stress on Bar 4 (4-5)	[-119.07 MPa, -0.025 MPa]
Stress on Bar 5 (5-6)	[-129.29 MPa, -64.65 MPa]
Stress on Bar 6 (6-1)	[-64.65 MPa, -0.010 MPa]
Stress on Bar 7 (2-5)	[-119.07 MPa, -0.0089 MPa]
Stress on Bar 8 (1-5)	[-155.33 MPa, -51.78 MPa]
Stress on Bar 9 (3-5)	[-310.66 MPa, -103.56 MPa]

Using APDL

APDL SCRIPT Steps

Description

- Preprocessing
 - Engineering Data: adjusting real constants and material properties
 - Model: adding nodal locations, element connections, nodal constraints, and loads
- Postprocessing
 - Plotting model
 - Listing nodal displacements, element stresses, element forces.

```

/filnam, PROBLEM_5
/title, PROBLEM_5
/prep7

!elements
et,1,link180      ! Element type no. 1 is LINK1

!real constants
r,1,78.54*5      ! Real constant number 1 (cross sectional area)

!material properties
mp,ex,1,1e6      ! Material Properties: E for material no. 1

!nodal locations
n,1,0,0,0        ! Nodal location 1
n,2,600*5,0,0    ! Nodal location 2
n,3,1200*5,0,0   ! Nodal location 3
n,4,1200*5,400*5,0 ! Nodal location 4
n,5,600*5,400*5,0 ! Nodal location 5
n,6,0,400*5,0    ! Nodal location 5

!element connections
real,1
en,1,1,2         ! Element 1 connects nodes 1 & 2
en,2,2,3         ! Element 2 connects nodes 2 & 3
en,3,3,4         ! Element 2 connects nodes 3 & 4
en,4,4,5         ! Element 2 connects nodes 4 & 5
en,5,5,6         ! Element 2 connects nodes 5 & 6
en,6,6,1         ! Element 2 connects nodes 6 & 1
en,7,2,5         ! Element 2 connects nodes 2 & 5
en,8,1,5         ! Element 2 connects nodes 1 & 5
en,9,3,5         ! Element 2 connects nodes 3 & 5

!nodal constraints
d,1,ux,0         ! node 1 allowed no displacement in x-dir
d,1,uy,0         ! node 1 allowed no displacement in y-dir
d,3,ux,0         ! node 3 allowed no displacement in x-dir
d,3,uy,0         ! node 3 allowed no displacement in y-dir

```

APDL SCRIPT

```
!loading
f,6,fx,10000*5      ! force on node 2 in x-dir is 10000
f,5,fy,-20000*5     ! force on node 2 in y-dir is 20000

finish

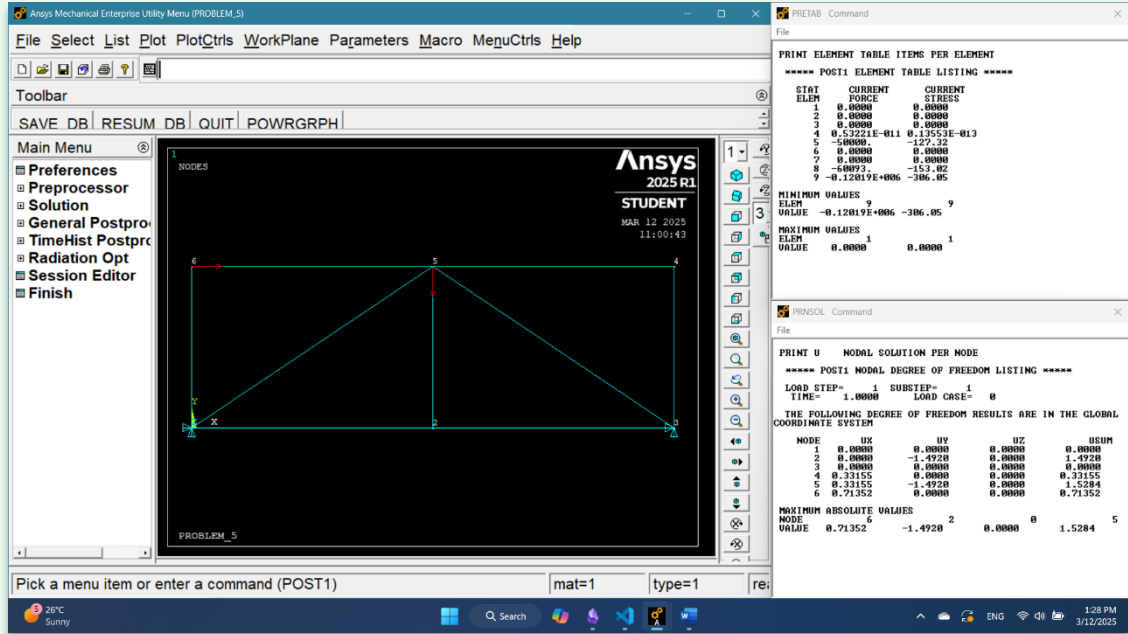
/solu               ! select static load solution
antype,static
solve
save
finish

/post1              ! start the post processor

etable,Force,SMISC,1    ! create table for element axial forces
etable,Stress,LS,1      ! create table for element stress values

/output,resultsfile,dat ! define a file, resultsfile.dat, in the working
directory

prnsol,u            !write the nodal solution to file
pretab,Force,Stress
```



NODE	UX (mm)	UY (mm)	UZ (mm)	USUM (mm)
1	0	0	0	0
2	0	-1.492	0	1.492
3	0	0	0	0
4	0.33155	0	0	0.33155
5	0.33155	-1.492	0	1.5284
6	0.71352	0	0	0.71352

Drive Link

https://drive.google.com/drive/folders/1FkCf3xJ__URDsqu3UmqLUp3LnrJgKhGPr